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Overview of effects of motor learning strategies in neurological and geriatric populations: a systematic mapping review

Running head: Motor learning strategies in neurological and geriatric populations

Li-Juan Jie PhD,[1*] Melanie Kleynen PhD,[1*] Guus Rothuizen MSc,[1] Elmar Kal PhD,[2,3] Andreas Rothgangel PhD,[1] Susy Braun PhD[1]

[1] Research Centre for Nutrition, Lifestyle and Exercise, School of Physiotherapy, Zuyd University of Applied Sciences, Faculty of Health, Heerlen, The Netherlands

[2] College of Health, Medicine and Life Sciences, Brunel University London, UK

[3] Centre for Cognitive Neuroscience, Brunel University London, UK

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Corresponding author: Li-Juan Jie, Nieuw Eyckholt 300, 6419 DJ Heerlen, 045 400 6060, Li-Juan.Jie@zuyd.nl

* Li-Juan Jie and Melanie Kleynen have contributed equally to this work.

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Abstract

Introduction: Motor learning plays a central role in neurological and geriatric rehabilitation. The wide range of motor learning strategies and increase in evidence can make it difficult to make informed decisions about the use of motor learning strategies in practice. This review's aim was to provide a broad overview of the current state of research regarding the effects of seven commonly used motor learning strategies to improve functional tasks within older neurological and geriatric populations.

Method: A systematic mapping review of randomised controlled trials was conducted regarding the effectiveness of seven motor learning strategies – errorless learning, analogy learning, observational learning, trial-and-error learning, dual-task learning, discovery learning, and movement imagery – within the geriatric and neurological population. PubMed, CINAHL, and Embase databases were searched. The Risk of Bias 2 tool was used to assess bias; additionally, papers underwent screening for sample size justification.

Results: Eighty-seven articles were included. Identified articles regarding the effects of the targeted motor learning strategies started around the year 2000 and mainly emerged since 2010. Eight different populations were included, e.g. Parkinson's, and stroke. Studies were not equally balanced across the motor learning strategies or target groups and overall showed a moderate to high risk of bias. Positive trends regarding effects were observed for dual-tasking, observational learning and movement imagery.

Conclusions: The findings show a skewed distribution of studies across motor learning interventions, which have been researched within a variety of populations. Methodological shortcomings make it difficult to draw firm conclusions regarding the effectiveness of motor learning strategies. Future researchers are strongly advised to follow guidelines that aid in maintaining methodological quality. Moreover, alternative designs fitting the complex practice situation should be considered.

Keywords: Motor learning, systematic review, mapping review, errorless learning, analogy learning, observational learning, trial-and-error learning, dual-task learning, discovery learning, and movement imagery.

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Motor learning - defined as a relatively permanent change in performance or behaviour[1] - plays a central role in the rehabilitation of neurological and geriatric rehabilitation.[2-4] Healthcare professionals, such as physical and occupational therapists, support patients to acquire or relearn a broad range of different motor skills, e.g. walking or reaching, to help them regain independence in activities of daily living (ADLs). [1-4] Certain general principles of skill training, like the frequency and specificity of practice and number of repetitions, are now widely recognised as being crucial to effective rehabilitation. In the last two decades, evidence has accumulated to suggest that *how* skills are taught may also be of relevance, [5,6] and guidelines now recommend incorporating motor learning strategies (e.g. implicit and explicit motor learning) into treatment approaches to improve rehabilitation success.[7] However, in many cases, there remains a lack of clear guidance on how different motor learning strategies can best be incorporated (and which can best be used for whom).[7-9] The vast variety of motor learning strategies from which health care professionals can choose, together with the rapid growth in publications and lack of overview of the effectiveness of these different strategies make it challenging to make informed decisions regarding the appropriate treatment approach and motor learning strategies. Further, many healthcare professionals seem to acquire novel knowledge unsystematically and in a fragmented manner.[9,10]

In order to support clinicians' decision-making and aid evidence-based implementation of motor learning strategies in their clinical practice, Kleynen and colleagues[11] developed a practical framework based on the broad distinction between conscious and non-conscious attributes of the motor learning process. This distinction proposes that implicit motor learning targets more non-conscious attributes of the motor learning process, whereas explicit motor learning targets more conscious attributes of the motor learning process.[12,13] The framework includes seven common motor learning strategies, which have been categorised as promoting more implicit or explicit motor learning: errorless learning, dual-task learning, analogy learning, discovery learning, observational learning, movement imagery, and trial-and-error learning.[11,14] The framework was informed by practice-based evidence from experts in different fields (e.g. researchers, health care professionals) as well as by research results that underpin these different learning strategies' working mechanisms. Currently, most evidence regarding the effectiveness of more implicit and explicit forms of motor learning is based on studies using laboratory tasks, e.g. Kal et al.[15] To support healthcare professionals in making informed decisions

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about the use of the seven motor learning strategies, more insight into their effectiveness in functional tasks is needed.

Systematic reviews potentially provide therapists with an accessible overview of available evidence to support their decision-making process. However, these reviews are often limited to a single motor learning strategy (e.g. errorless learning) within specific target populations (e.g. pathology- or disease-based), and focus on a single measurement outcome as to allow data pooling or synthesis (e.g., see[16-18] for excellent examples). In clinical practice, however, therapists treat various populations with a great variety of motor problems (and thus outcomes), rehabilitation needs, and preferences. Therapists therefore may need to switch between strategies, both within and between patients, to provide an optimal learning environment – but lack clear guidance to base this on as more comprehensive overview of the motor learning literature is lacking. This study's aim was to perform a systematic mapping review of randomised controlled trials (RCTs) to provide a comprehensive overview of the effects across the seven motor learning strategies incorporated in Kleynen et al.'s motor learning framework[11] for neurological and geriatric populations. In addition, the content of the interventions is described to gain more insight into how therapists could perform the different strategies in clinical practice.

Methods

This systematic review was conducted in two parts. The first part consisted of a quantitative analysis and focused on mapping the included studies to gain a quick overview of how many were published per (sub)population and per motor learning strategy over time. The second part included a descriptive analysis of motor learning intervention contents and effects, critically appraised in light of the studies' risk of bias and sample size justification.

Eligibility criteria

The population included all adults older than 60 and was not restricted to certain disorders. However, to optimise the search strategy, potentially relevant populations were specifically included in the search function (see search strategy). To ensure that the included studies would have direct clinical relevance, these studies' aim should be a performance improvement in a functional movement task.

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We defined a functional task as an activity that individuals perform as part of their daily routine, work, hobbies, or rehabilitation program. The control intervention group (comparator) was not predefined. The eligibility criteria for the selection of studies are presented in Table 1.

*** Table 1, about here ***

Search strategy

Two researchers (LJ, GR) searched the databases PubMed, CINAHL, and Embase for randomised controlled trials using the following combination of key search terms: ageing (older adults) OR neurological diseases (stroke OR parkinson OR dementia) AND motor learning strategies (analogy learning OR errorless learning OR trial and error OR discovery learning OR dual-task learning OR action observation OR mental practice) AND Activities of Daily Living (functional tasks). A detailed overview of the search strategy and the search terms used can be found in Appendix A. Additionally, reference tracking of the included studies was performed to identify additional studies.

Study selection process

Identification and screening of studies

Two researchers (GR, LJ) independently screened all retrieved articles from the databases based on the title, abstract, and keywords. After screening, the same two researchers obtained and assessed the full text of eligible articles independently according to the predefined selection criteria. In case of persistent disagreement, a third reviewer (MK) was consulted to reach a consensus.

Risk of bias assessment and sample size justification

The Cochrane Risk of Bias 2 (RoB2; [19]) was used to evaluate five different domains of bias, namely, randomisation, deviations from the intended intervention, missing outcome data, measurements of the outcome, and reporting of results. Based on specific criteria for each domain, an overall risk of bias was determined for every study, ranging from low risk of bias (green) to some concerns (yellow) to high risk of bias (red). Given the large number of studies included, a total of seven assessors (LJ, MK, GR, AR, EK, SB, RS) were involved in rating the risk of bias. To increase the reliability of the ratings, four calibration sessions were organised in which each item was discussed and further specified for

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the context of this study. Each article was assessed by two independent assessors. In case of disagreements, a third researcher was consulted. Authors of the included studies were not contacted to retrieve missing information. In addition to the standard RoB2 items, one extra item regarding the studies' sample sizes was added, since appropriate sample size justifications are often lacking or not transparently described.[20] Studies were also evaluated based on whether an a priori power analysis or other form of sample size justification was described. If sample size justification was described and achieved, this was categorised as 'green'. If no appropriate size justification was provided or the required sample size was not achieved, this was categorised as 'red'.

Data extraction

The following characteristics were extracted: year of publication, author, number of participants in total and per intervention/control group, population studied, gender, mean age, movement task trained, type of motor learning strategy/intervention, frequency and total duration of supervised practice, movement performance measures, assessment time points, and between-group effects.

Data analyses and synthesis

The analysis was divided into 1) a quantitative analysis in which the current available studies were mapped (Q1-3) and 2) a descriptive analysis of the studies' characteristics, the content of the intervention, and synthesis of the potential effects (Q4). For the quantitative analysis, all eligible articles were included. In the descriptive analysis, to increase the reliability of this study's conclusion regarding intervention effects, a second selection took place in which studies that scored 'high' on RoB2 and lacked (or failed to meet) an appropriate sample size justification were excluded.

As part of the quantitative analysis, a flowchart was presented to visualise the search and selection procedure. Further, the number of included studies per learning strategy over time and the type of patient population per learning strategy were mapped. An overview table per learning strategy was created presenting the risk of bias (low, some concerns, or high), sample size justification (yes/no), population, number of participants, and task trained.

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As part of the descriptive analysis, more in-depth information was provided regarding the population (type, group sizes, gender, age), intervention (motor learning strategy(ies), control intervention(s)), duration and frequency, task trained, movement performance measurement, and measurement moments (e.g. immediately after the intervention and, if applicable, also at follow-up). In the last step, we descriptively synthesised between group differences, both in terms of significance and direction of effects.

Results

Study selection

The study flow is visualised in Figure 1. In total, 2099 articles were identified. After deleting duplicates and screening the titles and abstracts, 236 articles remained, of which five articles could not be retrieved. The full text of 231 articles was obtained, and after screening, 90 were considered eligible and included for further analysis. Within this sample, there were three occasions in which the data of one single RCT was analysed in two different papers.[21-25] These papers were counted once, leading to a grand total of 87 studies included in the current review.

*** Figure 1, about here ***

Quantitative analysis (mapping)

In total, 87 studies were included. Six of the seven motor learning strategies were addressed (Figure 2). The most frequently described motor learning strategies were dual-task learning (n = 50 studies), mental practice (n = 19), and action observation (n = 12); no studies were found for discovery learning within these target populations. In total, eight different populations were identified within the included studies (Figure 2). Figure 3 visualises the number of studies published for each learning strategy over time.

*** Figures 2 and 3, about here ***

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An overview of the risk of bias scores, power, group size, tasks, and between-group differences is reported in Table 2. Of the 87 studies, 5 scored well on both the RoB (low) and sample size justification, while 18 had a high RoB and did not report a sample size justification. Included studies' group sizes ranged from 6 to 161 participants.

*** Table 2, about here ***

Descriptive analysis

Sixty-nine studies were left for the descriptive analyses, the results which are presented below per learning strategy.

Analogy learning

One study was included, with a sample size of 79 participants.[26] The task trained in the experimental group was walking in community-dwelling individuals after stroke. The practised gait parameters were chosen based on the patients' preferences and needs, as well as the clinical expertise of the therapists involved in the trial. The analogy learning instructions were personalised based on the individual's walking impairment and preferences. The effectiveness of analogy learning was compared to an explicit motor learning intervention. No between-group differences were observed either post-intervention or at the follow-up (low RoB, appropriate sample size justification). The intervention's duration was 3 weeks. The total intensity of training (i.e. the number of sessions multiplied by the duration of each session) was 270 minutes over 9 sessions. See Table 3 for more details.

*** Table 3 about here ***

Errorless learning and trial & error

Three studies were included, with sample sizes ranging from 30 to 161. Two studies included persons with Alzheimer's, of which one study trained ADL activities that were based on the patients' preferences and needs,[28] and the other practised a functional arm-hand task from the Action Programme test.[29] Furthermore, one other study included participants with transtibial amputations and

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246 trained the right technique for fitting a prosthetic limb.[30] The effectiveness of errorless learning was
247 examined in comparison to trial and error (n = 3).
248
249 The errorless learning intervention was structured in different ways. Frequent feed-forward
250 instructions (i.e. 'how to do') were provided before initiation of the task. To minimise mistakes, some
251 studies also provided cues verbally or pictorially (e.g. [29,31]). In contrast to the errorless learning
252 intervention, in the trial-and-error studies, participants were allowed to make mistakes and self-correct
253 their performance. In some studies, open-ended questions about the task were posed to participants
254 when repeated mistakes were observed (e.g. [28]).
255
256 One study[29] observed between-group differences in favour of the errorless learning intervention at the
257 follow-up measurement; however, there are some concerns about RoB and a lack of sample size
258 justification. Another study[30] observed between-group differences post-intervention (some concerns
259 about RoB, appropriate sample size justification). One study[28] did not observe any between-group
260 differences (low RoB, proposed sample size was not met in intention-to-treat analyses).
261
262 The total study duration of these interventions ranged from one single session[29,30] up to 10 weeks.[28]
263 The total intensity of training (i.e. the number of sessions multiplied by the duration of each session)
264 ranged from 15-30 minutes in one session[30] to 540 minutes over 10 training sessions.[28] Kessels and
265 Olde Hensken[29] did not specify the amount of time spent in each session. See Table 4 for more
266 details.
267
268
269 *** Table 4 about here ***
270
271 *Mental practice*
272 Eleven studies were included with sample sizes ranging from 11 to 121 participants. Two studies
273 included people with Parkinson's, both focusing on gait.[33,34] One study included participants after total
274 knee arthroplasty and practised knee extension.[35] Eight studies included people after stroke, two of
275 which focusing on gait,[36,37] three on upper limb activities,[38-40] two on daily life activities,[41-43] and one

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on sit-to-stand transfers.[44] The effectiveness of mental practice was examined in comparison to relaxation (n = 2), care as usual (n = 5), cognitive exercises (or mental rehearsal; n = 3), and standardised activities for the upper limbs (n = 1), and two studies specifically described that the intervention included a demonstration-then-practice element.

Mental practice interventions were often based on standardised protocols, scripts, or frameworks. The different stages of the mental practice intervention often included familiarisation with the task (e.g. analysis of the task sequence) and mental practice aspects (e.g. kinematic components), followed by internal imagery, mental rehearsal, and overt task performance. Two studies used audio instructions, while in one other study, the mental practice intervention was guided by a computer program. Different types of mental practice reported in the studies included kinaesthetic, visual, and motivational imagery.

One study[42] observed between-group differences in favour of the intervention post-intervention and at the follow-up measurement (some concerns about RoB, sample size justification lacking). Five studies[34,35,39,43,44] observed between-group differences in favour of the mental practice intervention post-intervention (RoB ranged from some concerns to high, one with appropriate sample size justification). Three studies[33,38,41] did not find any between-group differences (all with some concerns about RoB, two with appropriate sample size justification), and one study[36] did not calculate any between-group differences.

The total study duration of these interventions ranged from 3 weeks[36 42,43] up to 6 weeks.[39] The total intensity of training (i.e. the number of sessions multiplied by the duration of each session) ranged from 180 minutes over 12 sessions[36] to 900 minutes over 15 training sessions.[42,43] Braun et al.[33,41] did not specify the amount of time spent in each session. See Table 5 for more details.

*** Table 5 about here ***

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Observational learning

Eleven included studies had sample sizes ranging from 18 to 102 participants. Three studies included people after stroke who only practised upper limb activities.[51-53] Five studies included people with Parkinson's, focusing on gait in general (n = 1)[54] or freezing of gait in particular (n = 4).[25,55-57] Two studies included orthopaedic patients in which daily activities,[58] mobilisation exercises, and transfers[59] were trained. One study included older adults who practised walking.[60]

In all studies, observational learning was combined with or integrated into different types of functional training. In all studies, observational learning was applied through watching short movies of the task, exercise, or strategy to be learned. One study used videos that were composed of images and sounds (sonification).[25] Most studies (n = 9) investigated the effects of observing functional movements in comparison to observation of landscape videos or abstract pictures.[51,53-60] Two studies compared observational learning to a functional training intervention without (action) observation.[52,61]

Four studies[51,53,60,61] observed between-group differences in favour of the action observation intervention post-intervention and at the follow-up measurement (all with some concerns about RoB, three with appropriate sample size justification). Five studies[52,55,56,58,59] observed between-group differences in favour of the intervention at either of the measurement points (all with some concerns about RoB, two with appropriate sample size justification). One study[54] did not find any between-group differences (some concerns about RoB, appropriate sample size justification), and one study[57] did not calculate any between-group differences.

The total study duration of these interventions ranged from 8 days[54] up to 8 weeks.[61] The total intensity of training (i.e. the number of sessions multiplied by the duration of each session) ranged from 432 minutes over 18 sessions[58] to 960 minutes over 16 training sessions.[61] Jaywant et al.[54] did not specify the amount of time spent in each session. See Table 6 for more details.

*** Table 6 about here ***

Dual-task learning

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Forty-three studies were included, with a sample size ranging from 12 to 134. Three included persons after stroke, of which two studies[62,63] practised walking and one study[64] practised daily life activities. Three studies included orthopaedic patients, who all practised balance exercises.[65-68] Five studies included persons with dementia, and all practised walking and/or balance tasks.[69-73] Six studies included people with Parkinson's, of which two studies focused on balance training,[74,75] three practised walking,[76-78] and one practised aquatic exercises.[21] Seventeen studies included older adults, of which 13 practised walking and/or balance exercises[79-91] and the other studies practised aerobic exercises,[92] stepping exercises,[93] resistance training,[94] stationary biking,[95] or agility and strength training.[96] Three studies included older adults with balance impairments and practised walking and/or balance.[24,96-98] One study included older adults with fall histories and practised walking and balance.[99] Three studies included older adults with cognitive impairments and practised walking and/or balance[100,101] or dancing.[102]

Within the dual-task conditions, the secondary task was either a motor or a cognitive task. Secondary motor tasks included (avoiding) obstacles, carrying or playing with obstacles (e.g. a grocery bag, a tray, rattle, umbrella, or musical instruments), exercises with a ball (e.g. bouncing, passing, throwing, catching, holding, kicking) practicing daily life activities such as (un)buttoning a shirt, putting beans in a container (non-dominant hand) and unscrewing a nut and bolt, drawing a letter on the floor with one of their feet. Secondary cognitive tasks included engaging in conversations; singing; arithmetic tasks (e.g. 2-forward and 3-backward calculations); repeating animals' names; reading words or sentences backwards; counting/reciting the days of week; simple word games (e.g. coming up with a word that starts with the last letter of the previous word or naming as many words starting with the letter P (or another random letter); remembering cards; repeating phrases; playing phonemic word chain games; reciting a poem; answering questions about the participants' orientation to a person (identifying their name), time (date, month, or year), and place (current location); reacting to virtual situations (e.g. you're in a taxi but do not have your wallet); explaining the order of wearing clothes (e.g. dress, skirt, shirt, tie); talking about daily routines; making a shopping list; categorisation (e.g. types of land animals, drinks, colours, objects, boys' and girls' names, flowers, vegetables, fruit); clock face task; alternative uses (e.g. name an object and come up with alternative uses for that object); a creativity task (e.g. name as many objects that you know that are tall); letter fluency task; planning; singing a

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song; comparing drawings and naming differences; word spelling (fast as possible); auditory Stroop task; remembering shapes and colours; responding to auditory cues (fast as possible); and paying attention to tripping hazards. Finally, secondary tasks presented through VR games, e.g. playing a ball game, reactive boxing game, or cleaning windows.

Eight studies observed between-group differences[70,71,77,94,101,103,104] in favour of the dual-task learning intervention post-intervention and at the follow-up measurement (two studies with low RoB, six with some concerns about RoB, three with appropriate sample size justification). Nineteen studies[24,64-66,72-74,78,81,83,86-90,97-99,105] observed between-group differences only immediately after the intervention (two with low RoB, 13 with some concerns about RoB, four with high RoB, 13 with appropriate sample size justifications). Eleven studies[62,63,67,69,75,79,80,84,85,93,95] did not observe any between-group differences (all with some concerns about RoB, three with appropriate sample size justifications). Five studies[76,96,100,102,106] did not report the between-group effects (one with low RoB, four with some concerns about RoB, three with appropriate sample size justifications).

The total study duration of these interventions ranged from 1 day[83] up to 26 weeks.[87,92] The total intensity of training (i.e. the number of sessions multiplied by the duration of each session) ranged from 40 minutes in one session[83] to 4875 minutes over 65 training sessions.[92] See Table 7 for more details.

*** Table 7 about here ***

Discussion

This systematic mapping review provides a broad overview of the available studies on seven motor learning strategies, including their effect on improving functional tasks in neurological and geriatric populations. In total, 87 studies were identified, covering six of the seven included motor learning strategies. The most frequently researched motor learning strategies were dual-task learning (n = 50

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394 studies), mental practice (n = 19), and action observation (n = 12). No studies were found for
395 discovery learning.
396
397 ***Overview of available studies***
398 Mapping of the publications gave more insight into the development of publications over time and the
399 distribution and quality of RCTs regarding the seven strategies and different target groups. In the early
400 1990s, new scientific insights were published regarding recovery mechanisms and neuroplasticity of
401 the brain, which fuelled interest in the potential role of motor learning in rehabilitation.[107] Research on
402 the effects of motor learning strategies to improve functional tasks started around the year 2000, with
403 studies on mental practice and publications increasing substantially from 2010 on. In general, there
404 seems to be quite a big delay between the discovery of a learning strategy's (potential) working
405 mechanisms and its evaluation in applied research through RCTs. For example, the mirror-neuron
406 system and its role in our ability to learn by imitating others was discovered in the early 1990s,[108,109]
407 but it took about 20 years before the effects were evaluated within rehabilitation. Comparably, Mellit
408 and Petit[110] showed through fMRI that during imagery and performance of a motor skill, almost the
409 same brain areas are active, which gave a huge impulse for research on movement imagery in sports.
410 But it wasn't until 2007 that an increase in studies within (neuro)rehabilitation was seen.
411
412 Based on the results of our Delphi study[111], we expected the number of included studies to be
413 (somewhat) equally balanced across the motor learning strategies. After all, a substantial number of
414 experts had identified these seven motor learning strategies as the 'most used and well-known'
415 strategies within their field. This was, however, not the case. By far, dual-task learning has been
416 examined the most. There are several reasons which may explain why. First, dual-tasking is highly
417 prevalent in daily life activities and needs to be practised as such in every context (specificity).[112]
418 Additionally, some of the first clinical studies conducted showed very promising results (e.g.[23,24,68])
419 which might have led to an increase in similar, repetitive research paradigms: in 38 of the 50 included
420 studies, walking or balance was trained using a secondary cognitive task. Interestingly, no studies on
421 the use of discovery learning were included (one was excluded during the screening process). And
422 only two studies on analogy learning fulfilled our inclusion criteria.[26,27] This may be explained by the
423 fact that analogy learning is a relatively new concept, which was first translated to rehabilitation in

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2014 for persons after stroke[113] and with Parkinson's disease.[26,27,114] Trial-and-error learning was not researched as the experimental intervention in the included studies but was only used as a control condition for errorless learning in four studies. This is contrary to other fields of research (e.g. children, sports), in which trial-and-error learning and discovery learning have been assessed more extensively.[115]

Although the target population was defined broadly in our search and inclusion criteria, motor learning strategies were studied in only five populations. Older adults (without other specific motor or cognitive problems) were the most researched, followed by persons with Parkinson's disease and after stroke. No studies were found on other neurological diseases such as multiple sclerosis, spinal cord injury, or traumatic brain injury. Within the included population, certain motor learning strategies appeared to be preferred over others. Errorless learning, for example, was almost exclusively researched in persons with dementia, dual-task learning was mostly studied in older adults, and mental practice was often used in studies to improve arm-hand ability in people after stroke. Based on clinical expertise these strategies may be seen as more suitable for these specific populations, but evidence for their effects is still unclear.

A critical finding of this review is the limited methodological rigour observed across the 87 included studies. Specifically, only five studies (6%) had a low risk of bias and justified *and* achieved their desired sample size. Most of the studies scored 'some concerns' (54 studies; 62%). The lack of a strong methodological foundation in many studies makes it difficult to reliably identify the true intervention effects.[116] We also noted that the interpretation of individual study findings was frequently inaccurate (e.g. solely reported within-group differences and/or interaction effects without reporting between-group differences). There seems to be a more general problem of regulating the risk of bias and ensuring the accuracy of the reporting within motor learning research.[117,118]

Effects of the seven motor learning strategies

Only four (5% of total) studies were deemed reliable enough to interpret effects reported, based on their RoB2 (category green; low RoB), sample size justification (category green), and reporting of between-group differences: Geroin et al.[77] found between-group differences for dual-task gait and

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functional training in persons with Parkinson's, Trombetti et al.[87] and Javadpour et al.[81] found between-group differences for dual-task balance training in older adults, and Jie et al.[26] did not find significant between-group differences in gait for analogy learning in persons after stroke. Within the remaining moderately reliable studies, there seems to be a clear positive trend favouring observational learning across various populations (8/11 studies), when applied in addition to care as usual. Within mental practice (8/19 studies) and dual-task learning (27/43 studies), between-group effects favouring the intervention group were found in about half and the majority of the included studies, respectively. For dual-task learning, some trends may be observed within populations, as between-group effects in favour of the dual-task interventions were significant for the majority of the studies in persons with Parkinson's (5/6 studies), dementia (4/5 studies), and older adults with balance impairments (3/3 studies). Remarkably, two studies in dual-task learning in persons with dementia[70] and older adults with balance impairments,[98] found between-group differences in favour of the single-task (control) intervention.

Study limitations

Our search strategy was carefully prepared by experts in the field of literature review and motor learning. Still, we might have missed studies due to limitations in the search strategy (e.g. specific included search terms) and the categorisation of studies within databases. Publication bias might influence our findings; given the diversity of included studies, a funnel plot was not feasible.

As with any review article, our conclusions are subject to some common points of criticism concerning the standardised assessment of the studies' quality (Minozzi et al.[119]). Despite careful preparation, we still experienced that the use of the RoB2 tool left room for interpretation and needed additional effort to increase the reliability of the assessment. However, missing information in the texts was not retrieved by contacting the authors. Information not reported is not necessarily information not retrieved, and therefore, the criteria list assesses the study's report, not necessarily the quality of the study.

In line with earlier reviews, we decided to include an additional criterion (i.e. sample size justification) in our assessment of the included studies. An absence of sample size justification is not inherently

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problematic but increases uncertainties when evaluating effects.[116] Therefore, we excluded studies with a high risk of bias and without sample size justification from the descriptive analyses.

Future research

There are several considerations for researchers when conducting applied clinical motor learning research. Researchers should consider evaluating potential rehabilitation strategies not only *within* but also *across* populations.[120] Despite the anatomic and pathophysiological differences of target populations, these groups share many similarities, ranging from comparable cellular and neuro-physiological responses and recovery mechanisms to the effects of training in motor learning.

We also think that we should reconsider whether RCTs are the best fit for assessing complex training interventions. When conducting RCTs, researchers need to choose between internal validity (e.g. controlled context in laboratory settings) vs external validity reflecting daily practice (e.g. more 'uncontrolled' context with potential biases).[121,122] This may (partly) explain the overall moderate to high risk of bias and perhaps the absence of effects on some occasions. Hence, researchers should consider different research designs, e.g. cohort studies or multiple baseline designs, which might be more suitable for pragmatic trials with complex interventions. To facilitate interpretations of study results by therapists, researchers should also consider using clinically relevant differences, e.g. referring to minimally clinically important differences (MCIDs).[123] Likewise, to further increase the transferability to clinical practice, careful attention should be given to description of the interventions. The TIDieR (Template for Intervention Description and Replication) can be used as a checklist and guide to ensure interventions are reported with sufficient detail.[124]

Conclusions

The results of this study provide an overview of the current state of evidence regarding seven motor learning interventions in older neurologic and geriatric rehabilitation. The findings clearly show a skewed distribution of studies across motor learning interventions that have been researched within five target populations. The methodological shortcomings, e.g. high risk of bias and lack of appropriate sample size justifications, make it difficult to draw firm conclusions regarding the effectiveness of motor learning strategies. Hence, this review cannot provide a strong basis for

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514 therapists to rely on in their decision-making. Based on observed trends, therapists may consider (to
515 continue) using dual-task learning, observational learning and movement imagery. While waiting for
516 future research, therapists may also consider the other motor learning strategies based on their own
517 experiences and patients' preferences;
518 the description of the interventions of the included studies could be an example of how to apply
519 different strategies in daily practice within the different neurological and geriatric target populations.

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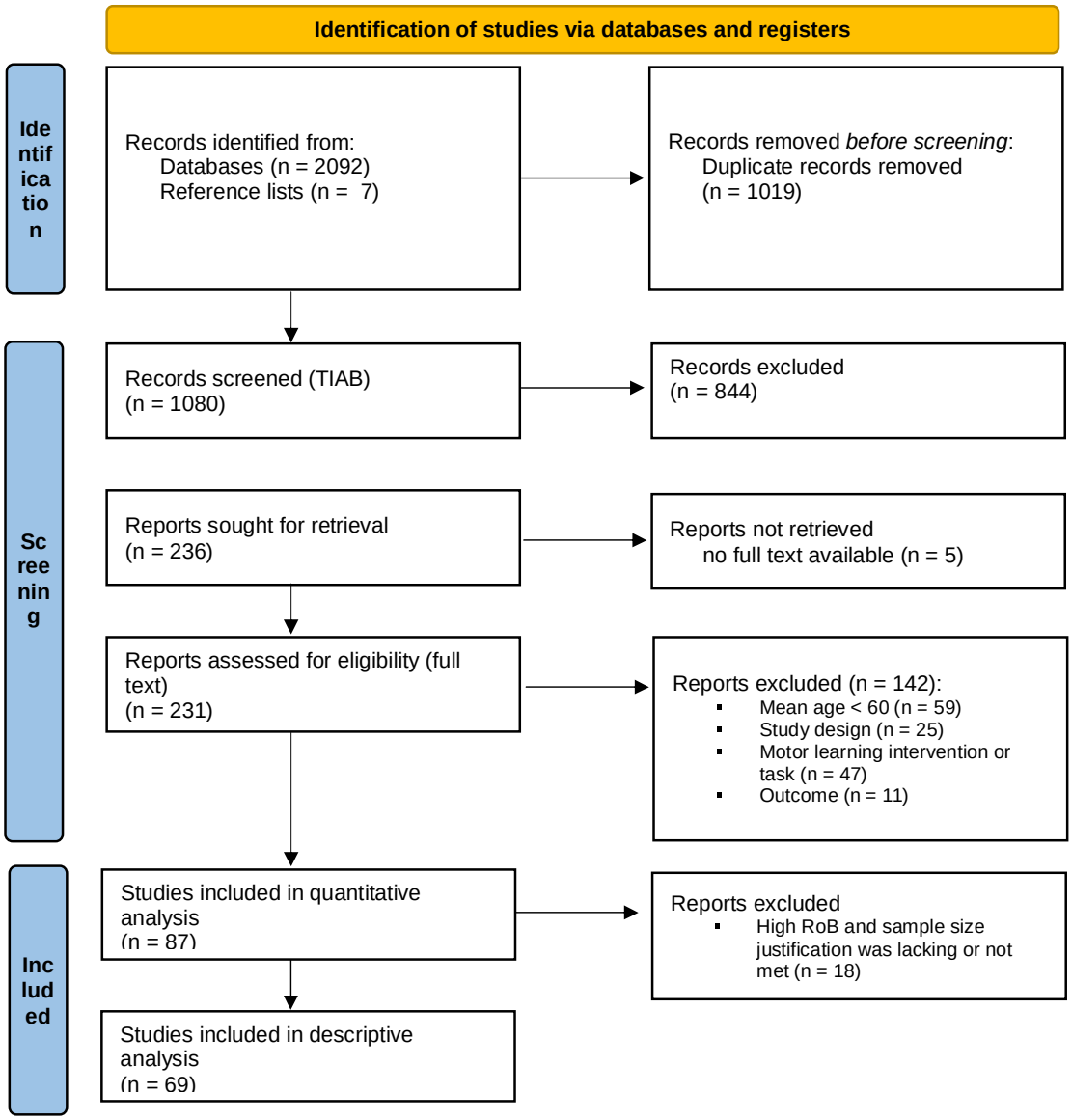
Figure 1. PRISMA flowchart for the inclusion of studies.

Figure 2. Visualisation of the number of studies identified per learning strategy including the sub-populations covered.

Figure 3. The year of publication per motor learning strategy. The identified motor learning strategies are based on the seven best-known and most-used motor learning strategies as embedded in the framework of Kleynen et al.[11] Data of six motor learning strategies were included, as no studies for discovery learning were identified.

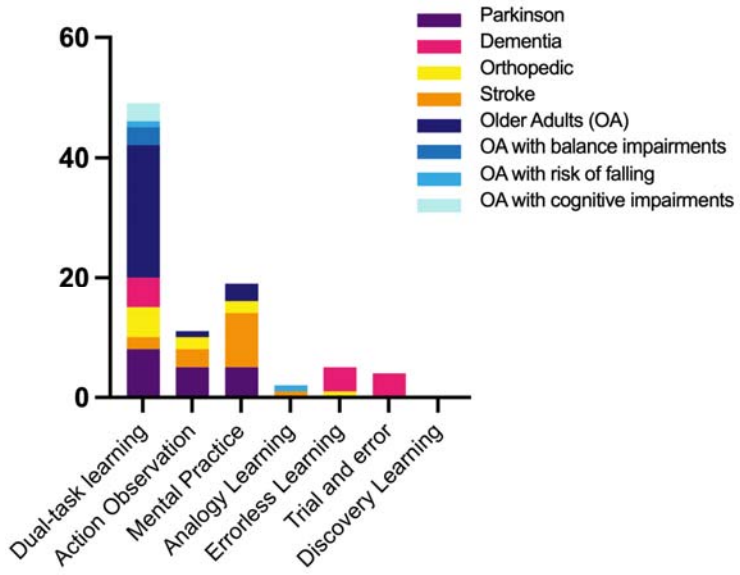
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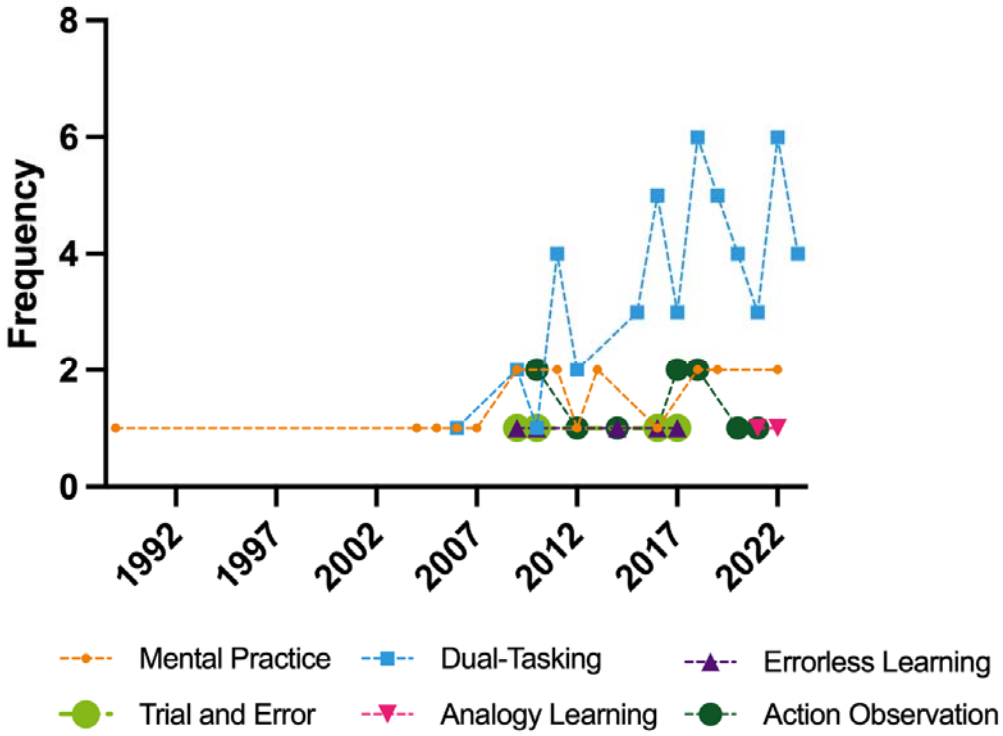
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Table 1. Overview of selection criteria.

	Inclusion	**Exclusion**
Population	Mean ≥ 60 years	Mean age < 60 years
Intervention	Errorless learning, analogy learning, observational learning, trial-and-error learning, dual-task learning, discovery learning, and mental imagery	Multimodal training interventions to ensure training effects are due to one of the seven interventions rather than other training modalities or combined interventions
	≥ 1 training session	< 1 training session
Outcome	Physical movement performance outcome assessed immediately after the intervention (acquisition) and/or at a delayed time point (retention/ transfer)	No motor performance outcome was measured, e.g. only magnetic resonance imaging or electroencephalogram. No serial reaction time outcomes
Study Design	Randomised controlled trial	Any other non-randomised trial design
Language	English, German, or Dutch	-
Accessibility	-	No full text available

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Table 2. Risk of bias scores, power, group size, task, and overall outcome of the included studies

Reference	Population	Overall RoB2	Power	Group size	Task[1]	Difference between groups	
Analogy learning							
Stroke	Jie et al., 2021	+	+	N = 79	Walking	Post: NS	Follow-up: NS
Older adults with risk of falling	Mak et al., 2022	-	-	N = 56	Walking	Post: NS	Follow-up: NS
Errorless learning							
Orthopaedic	Donaghey et al., 2010	!	+	N = 30	Arm-hand function ability	Post: S (2/5 outcomes)	Follow-up: NA
Alzheimer's	Bourgeois et al., 2016	-	-	N = 74	ADL	Post: NS	Follow-up: NS
	Thivierge et al., 2014	-	-	N = 17	ADL	Post: NS	Follow-up: NS
	Kessels et al., 2009	!	-	N = 60	Arm-hand function ability	Post: S (1/1 outcome)	Follow-up: S (1/1 outcome)
	Voigt-Radloff et al., 2017	+	-	N = 161	ADL	Post: NS	Follow-up: NS
Mental practice							
Parkinson's	Braun et al., 2011	!	+	N = 47	Walking and sit-to-stand	Post: NS	Follow-up: NS
	Silva et al., 2019	-	-	N = 18	Walking and balance	Post: NS	Follow-up: NA
	El-wishy et al., 2013	!	-	N = 26	Walking	Post: S (6/6 outcomes)	Follow-up: NA
	Marques de Melo Santiago et al., 2015	-	-	N = 20	Walking	Post: NS	Follow-up: NS
	Monteiro et al., 2018	-	-	N = 14	Mobility and balance	Post: S (3/3 outcomes)	Follow-up: NA
Older adults	Batson et al., 2007	-	-	N = 6	Dynamic exercises	Post: NS	Follow-up: NA
	Linden et al., 1989	-	-	N = 23	Walking balance	Post: NS	Follow-up: NA
	Nicholson et al., 2018	-	-	N = 30	Walking (obstacle course)	Post: NS	Follow-up: NA
Orthopaedic	Korbus & Schott., 2022	-	-	N = 29	Wrist movements	Post: NS	Follow-up: NA
	Paravlic et al., 2019	-	+	N = 34	Knee extension	Post: S (8/11 outcomes)	Follow-up: NA
Stroke	Braun et al., 2012	!	-	N = 36	Multiple functional tasks	Post: NS	Follow-up: NS
	Dickstein et al., 2013	!	+	N = 23	Walking	Post: NR	Follow-up: NR
	Guerra et al., 2022	-	+	N = 16	Walking, standing up	Post: NS	Follow-up: NA
	Ietswaart et al., 2011	!	+	N = 121	Upper limb activities	Post: NS	Follow-up: NA
	Liu et al., 2004	!	-	N = 46	ADL tasks	Post: S (3/5 outcomes)	Follow-up: S (1/1 outcomes)

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	Liu et al., 2009			N = 35	ADL tasks	Post: S (5/8 outcomes)	Follow-up: NA
	Malouin et al., 2009			N = 12	Standing up and sitting down	Post: S (2/2 outcomes)	Follow-up: NS
	Page et al., 2005			N = 11	Upper limb activities of daily living	Post: S (1/2 outcomes)	Follow-up: NA
	Park et al., 2016			N = 30	Upper limb activities through Wii games	Post: S (3/3 outcomes)	Follow-up: NA
Observational learning							
Orthopaedic	Bellelli et al., 2010			N = 60	Daily actions with the leg or trunk	Post: S (6/9 outcomes)	Follow-up: NA
	Villafañe et al., 2017			N = 31	Mobilisation exercises & transfers	Post: S (3/10 outcomes)	Follow-up: NA
Parkinson's	Agosta et al., 2017			N = 25	Strategies to circumvent FoG	Post: S (1/9 outcomes)	Follow-up: NS
	Jaywant et al., 2016			N = 23	Walking	Post: NS	Follow-up: NA
	Mezzarobba et al., 2018 & 2020			N = 22	Strategies to circumvent FoG	Post: S (5/19 outcomes)	Follow-up 1: S (8/19 outcomes) Follow-up 2: S (8/19 outcomes)
	Pelosin et al., 2010			N = 18	Strategies to circumvent FoG	Post: NS	Follow-up: S (2/6 outcomes)
	Pelosin et al., 2018			N = 64	Strategies to circumvent FoG	Post: NR	Follow-up: NR
Stroke	Franceschini et al., 2012			N = 102	Functional upper limb activities	Post: S (1/7 outcomes)	Follow-up: S (1/7 outcomes)
	Sale et al., 2014			N = 67	Functional upper limb activities	Post: S (4/6 outcomes)	Follow-up: S (4/6 outcomes)
	Mancuso et al., 2021			N = 36	Functional upper limb activities	Post: S (1/4 outcomes)	Follow-up: NA
Older adults	Rojasavastera et al., 2020			N = 33	Walking	Post: S (2/6 outcomes)	Follow-up: S (2/6)
Dual-task learning							
Stroke	An et al., 2021			N = 30	Activities of daily living	Post: S (1/1 outcomes)	Follow-up: NA
	Fishbein et al., 2019			N = 22	Gait training	Post: NS	Follow-up: NA
	Meester et al., 2019			*N* = 50	Walking	Post: NS	Follow-up: NA
Orthopaedic	Conradsson & Halvarsson, 2019			N = 68	Balance training programme	Post: S (10/32 outcomes)	Follow-up: NA
	Karagül et al., 2023			N = 43	Balance exercises	Post: S (4/10 outcomes)	Follow-up: NA
	Konak et al., 2016			N = 42	Static and dynamic balance	Post: S (3/6 outcomes)	Follow-up: NA

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	Uzunkulaoğlu et al., 2020	!	+	N = 50	Balance training programme	Post: NS	Follow-up: NA
	Vaillant et al., 2006	-	-	N = 56	Walking and balancing	Post: S (1/4 outcomes)	Follow-up: S (1/4 outcomes)
Dementia	Chen et al., 2018	!	-	N = 28	Walking	Post: NS	Follow-up: NS
	Ghadiri et al., 2022	!	+	N = 38	Walking and manipulative skills	Post: S (2/9 outcomes)[3]	Follow-up: S (2/9 outcomes)[3]
	Lemke et al., 2018	+	-	N = 105	Walking and balance	Post: S (25/55 outcomes)	Follow-up: S (10/55 outcomes)
	Menengiç et al., 2022	!	+	N = 20	Simple chair-based exercises	Post: S (4/6 outcomes)	Follow-up: NA
	Schwenk et al., 2010	!	-	N = 61	Walking and balance	Post: S (1/2 outcomes)	Follow-up: NA
Parkinson's	Fernandes et al., 2015	!	-	N = 15	Balance training	Post: S (2/5 outcomes)	Follow-up: NA
	Geroin et al., 2018	+	+	N = 121	Gait and functional training	Post: S (14/26 outcomes)	Follow-up: S (14/26 outcomes)
	Jäggi et al., 2023	!	+	N = 40	Balance and coordination	Post: NS	Follow-up: NA
	Park et al., 2021	-	-	N = 12	Drumming	Post: S (1/13 outcomes)	Follow-up: NA
	Silva et al., 2019, 2023	!	-	N = 25	Aquatic exercise	Post: S (4/6 outcomes), NR (2/6 outcomes)	Follow-up: S (4/6 outcomes), NR (2/6 outcomes)
	do Nascimento Silva et al., 2021	-	-	N = 10	Gait and balance training	Post: NS	Follow-up: NA
	Valenzuela et al., 2020	!	+	N = 40	Gait training	Post: S (11/20 outcomes)	Follow-up: Post: S (11/23 outcomes)
	Yang et al., 2019	!	+	N = 18	Gait training	Post: S (2/21 outcomes)	Follow-up: NA
Older adults	Brustio et al., 2017	!	+	N = 60	Balance and walking	Post: NS	Follow-up: NA
	Pessoa et al., 2020	!	-	N = 30	Walking and balance exercises	Post: S (3/6 outcomes)	Follow-up: NA
	Gregory et al., 2016	!	+	N = 44	Aerobic exercises	Post: S (3/6 outcomes)	Follow-up: S (3/6 outcomes)
	Hiyamizu et al., 2012	!	-	N = 36	Strength and balance training	Post: NS	Follow-up: NA
	Javadpour et al., 2022	+	+	N = 69	Balance training	Post E1 vs C: S (10/10 outcomes) Post E2 vs C: S (9/10 outcomes)	Follow-up: NA
	Kitazawa et al., 2015	!	-	N = 60	Step exercise programme	Post: NS	Follow-up: NA
	Castillo de Lima et al., 2023	-	-	N = 16	Agility training	Post: NS	Follow-up: NA
	Nascimento et al., 2023	!	+	N = 44	Walking and balancing	Post: NR	Follow-up: NA

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	Nematollahi et al., 2016	-	-	N = 44	Balancing	Post: NS	Follow-up: NA
	Norouzi et al., 2019	!	-	N = 60	Resistance training	Post E1 vs E2 & C: S (1/1 outcome)	Follow up E1 vs E2 & C: S (1/1 outcome)
	Plummer-D'Amato et al., 2012	!	-	N = 17	Walking and balancing	Post: NS	Follow-up: NA
	Raichlen et al., 2020	!	+	N = 74	Stationary biking	Post: NS	Follow-up: NA
	Sinaei et al., 2016	!	-	N = 24	Balance training	Post: NS	Follow-up: NA
	Tasvuran Horata et al., 2021	-	+	N = 32	Walking and balancing	Post: S (5/8 outcomes)	Follow-up: NA
	Trombetti et al., 2011	+	+	N = 134	Balance training	Post: S (8/17 outcomes)	Follow-up: NA[2]
	Uemura et al., 2012	!	-	N = 15	Agility and strength	Post: NR	Follow-up: NA
	Wollesen et al., 2015	!	-	N = 38	Walking and Balancing	Post: S (7/36 outcomes)	Follow-up: NA
	Wollesen et al., 2017A	!	+	N = 78	Walking and balancing	Post: S (8/22 outcomes)	Follow-up: NA
	Wollesen et al., 2017B	-	+	N = 95	Walking and balancing	Post: S (3/12 outcomes)	Follow-up: NA
	Yamada et al., 2011A	-	+	N = 84	DVD group training	Post: S (1/4 outcomes)	Follow-up: NA
	Yamada et al., 2011B	!	+	N = 53	DVD group training	Post: NR	Follow-up: NA
	You et al., 2009	-	-	N = 13	Gait training	Post: NR	Follow-up: NA
Older adults with balance impairments	Azadian et al., 2016	!	-	N = 30	Gait training and balancing	Post E1 & E2 vs C: S (5/13 outcomes) Post E2 vs E1 & C: S (3/13 outcomes)	Follow-up: NA
	Khan et al., 2018	!	-	N = 39	Balance training	Post: S (2/4 outcomes)[3]	Follow-up: NA
	Silsupadol et al., 2009A & B	!	+	N = 21	Balance training	Post E1 & E2 vs C: S (2/10 outcome) Post E1 vs E2 & C: S (1/10 outcome)	Follow-up: NA
Older adults with history of falls	Park et al., 2022	!	+	N = 58	Walking and balance training	Post: S (2/2 outcomes)	Follow-up: NA
Older adults with cognitive impairments	Delbroek et al., 2017	!	-	N = 20	BioRescue balance training	Post: NR	Follow-up: NA
	Kuo et al., 2022	!	+	N = 30	Walking	Post: E1 & E2 vs C: S (4/20 outcomes)	Follow-up: E2 vs C: S (4/20 outcomes)
	Parial et al., 2023	+	+	N = 60	Dancing	Post: NR	Follow-up: NR

C: control group; ADL: activities of daily living; FoG: Freezing of Gait; S: significant ($p < 0.05$); NS: not significant ($p > 0.05$); NR: not reported; NA: not applicable; [1] task trained in experimental condition; [2] cross-over trial; [3] effects in favour of control group

Table 3. Detailed study characteristics of one analogy learning study

STROKE				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Jie et al., 2021	N = 79 persons after stroke in chronic phase	*Task trained in experimental group:* Walking. The to-be-improved gait parameter was chosen based on the therapist's expertise and the participants' needs.		*Measurement instrument:* - 10 Meter Walk Test - Dual Task Costs motor task - Dual Task Costs cognitive task - Modified Dynamic Gait Index *Measurement moment:* Pre- and post-intervention and 4-week follow-up *Between-group differences:* No significant differences in any of the outcomes
	Analogy learning group: *n* = 38; 64.6 yrs (SD: 9.4)	A learning environment was created where participants were not (or minimally) aware of the underlying rules of the practised motor skill. Participants received *personalised analogy instructions* based on their individual walking problems and preferences.	9 sessions of 30 minutes in 3 weeks	
	Explicit learning group: *n* = 41; 67.8 yrs (SD: 11.6)	A learning environment was created so participants were very aware of the learning process. Participants received *detailed explicit instructions* on how to improve their gait performance.	9 sessions of 30 minutes in 3 weeks	

N = number of participants in total, n = number of participants per group, *SD = standard deviation, NA = not applicable, NR = not reported*

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Table 4. Detailed study characteristics of three studies on **errorless learning** and **trial and error**

ALZHEIMER'S				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Kessels et al., 2009	N = 60 persons with mild-to-moderate, severe dementia, or no dementia	*Task trained in experimental group*: Functional arm-hand tasks, e.g. removing a cork from a tube.		Measurement instruments: - Behavioural Assessment of the Dysexecutive Syndrome (BADS) *Measurement moments:* Pre- and post-intervention and 1-3 days follow-up Between-group differences: Difference in task performance on the BADS in favour of the errorless learning groups in persons with severe, mild-to-moderate, or no dementia.
	Errorless learning group: *Group 1 – severe dementia* *n* = 10; 83.6 yrs (SD: 8.1) *Group 2 – mild-to-moderate dementia* *n* = 10; 76.5 yrs (SD: 7.9) *Group 3 – controls: no dementia* *n* = 10; 72.7 yrs (SD: 11.0)	To prevent errors, participants received cues before completing a sequence of the to-be-completed task (e.g. 'You can use the hook to remove the lid').	1 session (duration of session NR)	
	Trial-and-error learning group: *Group 1 – severe dementia* *n* = 10; 83.2 yrs (SD: 7.1) *Group 2 – mild-to-moderate dementia* *n* = 10; 77.1 yrs (SD: 9.4) *Group 3 – controls: no dementia* *n* = 10; 71.9 yrs (SD: 8.9)	Initially, cues were not provided to complete the task. Cues were only provided (in the second instance) if they were unable to find and complete the correct next step, i.e. errors were made.	1 session (duration of session NR)	
Voigt-Radloff et al., 2017	N = 161 persons with Alzheimer's or mixed-type dementia	*Task trained in experimental group*: By shared decision-making, participants practised two tasks relevant to daily living.		Measurement instruments: Core Elements Method (CEM: a 7-point scale for each task where 1 = not performed at all as trained by the therapist; 7 = performed exactly as trained by the therapist) *Measurement moments:* Pre- and post-intervention, and 6- and 16-week follow-up Between-group differences: No significant differences in any of the outcomes.
	Errorless learning group: *n* = 81; 77.1 yrs (SD: 7.8)	The therapist divided the task into appropriate steps, demonstrated and (verbally) guided the participants through the steps. If potential errors occurred, then the therapist anticipated through providing a short demonstration of the correct performance, i.e. errors were prevented	9 sessions of 60 min in 10 weeks	
	Trial-and-error group: *n* = 80; 76.1 yrs (SD: 6.8)	Participants were asked to perform the task but did not receive instructions or demonstration. After 3 insufficient trials, open-ended questions were asked about the purpose of the task to find solutions.	9 sessions of 60 min in 10 weeks	

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Table 4 (continued). Detailed study characteristics of three studies on **errorless learning** and **trial and error**

ORTHOPAEDICS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Donaghey et al., 2010	N = 30 persons with unilateral transtibial amputations	*Task trained in experimental group*: Fitting a prosthetic limb.	All participants were first shown how to put on the prosthetic limb.	Measurement instruments: Video analyses of the final trial included: - Total number of correct steps - Number of omissions - Number of deviations - Number of hesitations - Time taken to complete the fitting sequence *Measurement moments:* Post-intervention Between-group differences: Difference in correct steps in the fitting sequence and number of errors in favour of the errorless learning group
	Errorless learning group: *n* = 15; 62 yrs (SD: 14.6)	The participants were 'talked through' the sequence of tasks, such that they were unable to make any errors. Participants received the appropriate limb parts needed to move correctly to the next stage of the fitting sequence.	1 session of 15-30 min	
	Trial-and-error group: *n* = 15; 66 yrs (SD: 6.8)	Learning proceeded via a trial-and-error process.	1 session of 15-30 min	

N = number of participants in total, n = number of participants per group, *SD = standard deviation, NA = not applicable, NR = not reported, BADS = Behavioural Assessment of the Dysexecutive Syndrome,*

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Table 5. Detailed study characteristics of 11 studies on **mental practice**

PARKINSON'S				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Braun et al., 2011	N = 47 persons with Parkinson's	*Task trained in experimental group:* Mobility tasks, e.g. walking and sit-to-stand exercises.		Measurement instruments: - Visual Analogue Scale on walking performance - Timed Up and Go - 10-Metre Walking Test Measuring moments: Pre- and post-intervention and 1-month follow-up *Between-group differences*: No significant differences in any of the outcomes.
	Mental practice group: *n* = 25; 70.0 yrs (SD: 8.0)	Usual care according to national guidelines and embedded mental practice, based on a pre-described 4-phase protocol (individual or group). Unguided mental practice was encouraged.	12 sessions of 30 min (or 6 sessions of 60 min) in 6 weeks	
	Control (relaxation) group: *n* = 22; 69.0 yrs (SD: 8.0)	Usual care according to national guidelines and embedded relaxation therapy, based on the principles of progressive muscle relaxation (individual or group). Unguided relaxation therapy was encouraged.	12 sessions of 30 min (or 6 sessions of 60 min) in 6 weeks	
El-wishy et al., 2013	N = 26 persons with Parkinson's	*Task trained in experimental group:* Mobility task and gait performance		Measurement instruments: - Gait speed - Step length - Hip excursion - Knee excursion - Ankle excursion - Functional Gait Assessment (FGA) Measuring moments: Pre- and post-intervention *Between-group differences*: Differences in gait speed and stride length in favour of the locomotor imagery training group. The ankle, knee, and hip excursions were larger in the imagery training group. Difference on the FGA in favour of the imagery training group.
	Locomotor imagery training group: *n* = 13; 71.0 yrs (SD: 4.2)	Protocolised physical therapy and locomotor imagery training. Participants watched 2 videos of a healthy older adult and patient walking (displayed from anterior, posterior, and side views). Motor imagery in accordance with a five-stage protocol: progressive relaxation, external imagery (analysis of task sequences), problem identification, internal imagery, and mental rehearsal.	12 sessions of 50-70 min in 4 weeks	
	Control (physical therapy) group: *n* = 13; 72.0 yrs (SD: 3.5)	Protocolised physical therapy training programme. Participants watched a documentary television programme on topics related to health for the same period of time.	12 sessions of 50-70 min in 4 weeks	

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Table 5. Detailed study characteristics of 11 studies on **mental practice** (continued)

ORTHOPAEDICS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Paravlic et al., 2019	N = 34 persons after knee arthroplasty	*Task trained in experimental group:* maximal voluntary isometric knee extension contraction.	Daily 45-minute continuous passive-motion session and various functional everyday movement exercises.	Measurement instruments: - Maximal isometric knee extension strength - 30-second chair stand test - Self-selected gait speed - Self-selected gait speed DT - Brisk-paced gait speed - Brisk-paced gait speed DT - Single support period - Double support period - Stride length - Cadence - Self-reported physical function Measuring moments: Pre- and post-intervention Between-group differences: Differences in maximal isometric knee extension strength, sit-to-stand performance, self-selected gait speed (in single- and dual-task conditions), and brisk-paced gait speed (in single- and dual-task conditions) in favour of the MIp.
	Movement imagery group (MIp): *n* = 17; 62.2 yrs (SD: 4.9)	Routine physical therapy plus movement imagery. MI was practised both during hospitalisation and at home. During the hospitalisation period, participants had been advised to imagine maximal voluntary isometric contractions. After discharge, at home, the MI was supported by an audio description (mp3 file) of the MI session.	20 sessions of 9 min in 4 weeks	
	Control group: *n* = 17; 60.0 yrs (SD: 5.7)	Routine physical therapy (no MI).	NA	

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Table 5. Detailed study characteristics of 11 studies on **mental practice** (continued)

STROKE				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Braun et al., 2012	N = 36 persons after stroke in the subacute phase	*Task trained in experimental group:* Functional tasks: walking, one self-chosen arm activity and one leg activity.	Multi-professional rehabilitation for 6 weeks.	Measurement instruments: - Numeric Rating Scale - Rivermead Mobility Index - Barthel Index - Berg Balance Scale - Motricity Index - Nine-Hole Peg Test - 10-Metre Walking Test Measuring moments: Pre- and post-intervention, 6-month follow-up Between-group differences: No significant differences in any of the outcomes.
	Mental practice intervention: *n* = 18; 77.7 yrs (SD: 7.2)	Mental practice based on a 4-stage framework embedded in therapy as usual. Participants were encouraged to practice unguided imagery training.	In some patients, between 10-20 minutes were added to the regular rehabilitation programme.	
	Control group: *n* = 18; 77.9 yrs (SD: 7.4)	Therapy as usual, and participants were encouraged to practice physically (unguided).	NR	
Dickstein et al., 2013	N = 23 persons after stroke in the chronic phase	*Task:* Walking tasks based on patients' goals.		Measurement instruments: - 10-Metre Walking Test - Falls-Efficacy Scale - Step activity monitor - Maximal activity (number of steps taken/min during the most active hour of the day Measuring moments: Pre- and post-intervention, 2-week follow-up Between-group differences: No significant differences in any of the outcomes.
	Integrated motor imagery: *n* = 12, 71.3 yrs (SD: NR)	Individual integrated motor imagery with three standardised imagery scripts (one/week) combining kinesthetics, visual imagery, and motivational imagery.	12 sessions of 15 min in 4 weeks	
	Control group (before cross-over): *n* = 11, 72.2 yrs (SD: NR)	Physical therapy for the affected upper extremity with three standardised types of exercises (one/week).	12 sessions of 15 min in 4 weeks	

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Table 5. Detailed study characteristics of 11 studies on **mental practice** (continued)

STROKE				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Guerra et al., 2022	N = 16 persons after stroke in the subacute phase	*Task trained in experimental group*: standing up from a chair and walking.	Sessions within both groups were followed by a physical practice session of 40 min.	Measurement instruments: - Timed Up and Go Test - Gait speed - Timed Up and Go Test – Assessment of Biomechanical Strategies - Muscle strength - Hip abduction (left and right) - Hip flexion and extension (left and right) - Knee flexion and extension (left and right) - Ankle dorsi- and plantarflexion (left and right) Measuring moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Mental practice group: *n* = 8; 61.5 yrs (range 59.5-65.5)	Mental practice based on a 5-stage protocol: physical practice, familiarisation with kinematic components, memorisation, relaxation, and mental practice.	12 sessions of 30 min in 4 weeks	
	Cognitive training group: *n* = 8; 66.5 yrs (range 57.0-74.5)	Cognitive exercise related to memorising, naming, and reasoning activities.	12 sessions of 30 min in 4 weeks	
Ietswaart et al., 2011	N = 121 persons after stroke (within 6 months post-stroke)	*Task trained in experimental group*: A variety of elementary movements, goal-directed movements, and activities of daily living of the upper limb.	All participants received physical therapy and normal care.	Measurement instruments: - Action Research Arm Test score - Grip strength - Manual dexterity performance speed - Barthel Index - Functional limitations profile Measuring moments: Pre- and post-intervention *Between-group differences*: No significant differences in any of the outcomes.
	Motor imagery training: *n* = 41; 69.3 yrs (SD: 10.8)	Motor imagery according to a standardised protocol which did not allow tasks to be individualised.	12 sessions of 45 min in 4 weeks	
	Attention-placebo control: *n* = 39; 68.6 yrs (SD: 16.3)	Training of mental rehearsal unrelated to motor control such as visual imagery of objects. Received the same amount of attention as the motor imagery training group.	12 sessions of 45 min in 4 weeks	
	Normal care: *n* = 42; 64.4 yrs (SD: 15.9)	Normal care with no additional training.	NA	

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Table 5. Detailed study characteristics of 11 studies on **mental practice** (continued)

STROKE				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Liu et al., 2004	N = 46 people after stroke (average 13.8 days after stroke)	*Task trained in experimental group*: ADL tasks incl. household, cooking, and shopping tasks.		Measurement instruments: - Average performance of 5 trained tasks (Week 1) - Average performance of 5 trained tasks (Week 2) - Average performance of 5 trained tasks (Week 3) - Average performance 5 untrained tasks (Week 3) - Average performance of 5 trained tasks (follow-up) - Fugl-Meyer Assessment Measure moments: Pre- and post-intervention, 1-month follow-up Between-group differences: Differences in performance assessments of the trained and untrained tasks in favour of the mental imagery group at Week 2, Week 3, and follow-up.
	Mental imagery: *n* = 26; 71.0 yrs (SD: 6.0)	Mental imagery-based intervention according to a 9-step protocol (task analysis, problem identification, task performance) guided by a computer program.	15 sessions of 60 min in 3 weeks	
	Functional retraining: *n* = 20; 72.7 yrs (SD: 9.4)	A functional retraining programme with a demonstration, then practice. Patients were required to practise the same tasks following a sequence and training schedule similar to that of the mental imagery programme.	15 sessions of 60 min in 3 weeks	
Liu et al., 2009	N = 33 persons in the acute phase of recovery	*Task trained in experimental group*: ADL tasks incl. household, cooking, and shopping tasks.	1-hour daily physical therapy – mobilisation, strengthening, and walking exercises	Measurement instruments: - Performance of 5 trained ADL tasks - Performance of 3 untrained ADL tasks Measure moments: Pre- and post-intervention Between-group differences: Differences were for 3 trained tasks (fry vegetables with meat, tidy table after meal, and go to park) and two untrained tasks (clean refrigerator, go to resource centre) in favour of the intervention group.
	Mental imagery: *n* = 16; 70.8 yrs (SD: 9.3)	Mental imagery programme including truncating the task (chunking), self-reflecting on deficits in performing it, feedback (using video playback), mentally rehearsing as if performing it (rehearsal), and actually carrying the task out.	15 sessions of 60 min in 3 weeks	
	Functional rehabilitation: *n* = 17; 69.7 yrs (SD: 7.4)	Conventional occupational therapy involving the demonstration and practice of the task under supervision.	15 sessions of 60 min in 3 weeks	

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Table 5. Detailed study characteristics of 11 studies on **mental practice** (continued)

STROKE				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Malouin et al., 2009	N = 12 persons after stroke in the chronic phase	*Task trained in experimental group*: Standing up from a chair and sitting down.		Measurement instruments: - Loading of the affected leg whilst rising. - Loading of the affected leg when sitting down. Measuring moments: Pre- and post-intervention, 3-week follow-up Between-group differences: Differences in limb loading whilst both standing up and sitting down in favour of the mental practice group post-intervention. No difference was found between the cognitive practice group and the group without training.
	Mental practice + physical practice: *n* = 5; 61.3 yrs (SD: 7.2)	A combination of mental practice with physical repetitions including preparation, instructions, mental repetitions, auto-estimation of motor imagery vividness, physical repetitions, and rest periods.	12 sessions of 60 min in 4 weeks	
	Cognitive practice + physical practice: *n* = 3; 61.0 yrs (SD: 8.5)	A combination of cognitive training (e.g. mental calculation, recall of numbers, questions about sports) with physical repetitions and rest periods.	12 sessions of 60 min in 4 weeks	
	No training: *n* = 4; 61.8 yrs (SD: 9.5)	No training.	NA	
Page et al., 2005	N = 11 persons after stroke in the chronic phase; 62.3 yrs (range, 53-71 yrs)	*Task trained in experimental group:* Functional movements of ADLs using affected arm, standardised for all patients of the experimental group.	All participants received therapy for the more affected arm for 12 sessions of 30 min in 6 weeks	Measurement instruments: - Motor Activity Log - Action Research Arm Test Measuring moments: Pre- and post-intervention Between-group differences: Differences in the Action Research Arm Test were found in favour of the mental practice group.
	Mental practice group: *n* = 6; (SD: NR)	Participants followed a pre-recorded audio mental practice intervention (audiotape) including relaxation, mental practice, and refocusing.	12 sessions of 30 min in 6 weeks	
	Control group: *n* = 5: (SD: NR)	Participants followed relaxation techniques according to a progressive relaxation protocol (audiotape).	12 sessions of 30 min in 6 weeks	

N = number of participants in total, n = number of participants per group, *SD = standard deviation, NA = not applicable, NR = not reported, FGA = Functional Gait Assessment*

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Table 6. Detailed study characteristics of eleven studies on **observational learning**

Reference	Population: Orthopaedic	Intervention and task	Amount of supervised practice	Measurement instruments, moments, and outcome
Bellelli et al., 2010	N = 60 persons after hip fracture or hip or knee replacement	Task: Daily actions with the leg or trunk.	Conventional post-orthopaedic rehabilitation programme for 1 hour a day, 6 days a week, for 3 weeks	Measurement instruments: - Functional Independence Measure (FIM) at discharge - FIM absolute functional gain (AFG) (change in scores from admission) - FIM absolute functional efficiency (AFG / length of stay) to discharge - FIM relative functional gain (AFG / (maximal possible FIM score – actual admission FIM) - Motor FIM subscore at discharge - Motor FIM subscore AFG - Tinetti at discharge - Tinetti AFG - Use of walking aids Measurement moments: Pre- and post-intervention Between-group differences: Between-group differences were found for the following outcomes in favour of the action observation group: FIM absolute functional gain, absolute functional efficiency, relative functional gain, Motor FIM subscore AFG, Tinetti AFG, and use of walking aids.
	Action observation group: *n* = 30; 71.9yrs (SD: 8.4)	Participants observed 3 short movies showing an actor performing daily actions with the leg or trunk (e.g. moving lower limbs, walking). Each short movie included 4 different 2-minute actions. After observation, participants performed the observed actions.	18 sessions of 24 min in 3 weeks	
	Control group: *n* = 30; 71.8yrs (SD: 6.9)	Participants observed short movies (for the same time as patients in the case group) showing scenes with no motor content (e.g. geographic documentaries). After observation, participants performed the same actions as in the experimental group.	18 sessions of 24 min in 3 weeks	
Villafañe et al., 2017	N=31 persons after primary hip replacement	Task: Mobilisation exercises and transfer practice.	Both groups: Continuous Passive Movements	Measurement instruments: - Pain - Active flexion - Active extension - Passive flexion - Passive extension - Barthel index - Short Form-36 – motor - Short Form-36 – mental - Tinetti scale - Lequesne Index Measurement moments: Pre- and post-intervention Between-group differences: Between-group differences were found for the following outcomes in favour of the action observation group: active flexion and extension and Tinetti.
	Action observation group: *n* = 14; 70.4 yrs (SD:7.5)	Participants watched a video of the exercise before performing independent exercise.	20 sessions (twice a day) of 30 min in 2 weeks	
	Control group: *n* = 17; 70.1 yrs (SD:7.7)	Participants watched a video nature scene before performing independent exercise.	20 sessions (twice a day) of 30 min in 2 weeks	

Table 6. Detailed study characteristics of eleven studies on **observational learning** (continued)

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Reference	Population	Intervention and task	Amount of supervised practice	Measurement instruments, moments, and outcome
Agosta et al 2017	N = 25 persons with PD	Task: Strategies useful in circumventing FOG.		Measurement instruments: - Unified Parkinson's Disease Rating Scale III (1x off, 1x on) - Freezing of Gait Questionnaire - Unified Parkinson's Disease Rating Scale II – FOG (1x off, 1x on) - Parkinson's Disease Questionnaire (PDQ-39) - Berg Balance Scale - 10-Metre Walking Test (1x comfortable, 1x speed) Measuring moments: Pre- and post-intervention, 4-week follow-up Between-group differences: Differences were found on the UPDRS III ON in favour of the AOT intervention.
	Action observation training (AOT): *n* = 12; 69.0 yrs (SD: 8.0 yrs)	Participants watched videos of strategies useful in circumventing freezing of gait (FOG) episodes; auditory cues were associated with the movements. Afterwards, patients were asked to imitate the observed actions to the beat of the auditory cues.	12 sessions of 60 min in 4 weeks	
	Landscape: *n* = 13; 64.0 yrs (SD: 7.0 yrs)	Performance of the same movements/actions used for the AOT group in the same order and amount of time with the same instructions, but video clips of landscapes were shown for the same length of time.	12 sessions of 60 min in 4 weeks	
Jaywant et al 2016	N = 23 persons with PD	Task: Watching and judging videos.		Measurement instruments: Measured in three conditions: straight line, walking with turns, dual-task - Walking speed (m/s) - Walking speed variability - Stride length (m) - Stride length variability - Gait asymmetry - Stride frequency (strides/s) - Stride frequency variability - Swing time (% of stride) - Swing time % variability *Home walking assessment (accelerometer)* - Stride frequency - Number of walking periods - Duration of each walking period Parkinson's Disease Questionnaire (PDQ-39) Measuring moments: Pre- and post-intervention Between-group differences: No between-group differences were found in any of the outcomes.
	Gait observation: *n* = 13; 63.7 yrs (SD: 6.2)	Participants viewed videos of actors with and without PD who were walking. They judged whether their walking in each video appeared healthy or resembled a PD-like gait pattern.	8 sessions, 1 session per day for 8 days	
	Landscape observation: *n* = 10; 65.8 yrs (SD: 8.7)	Participants viewed videos of landscapes with moving water in oceans, rivers, lakes, and waterfalls. Participants judged whether the water was moving 'roughly' or 'calmly'.	8 sessions, 1 session per day for 8 days	

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Table 6. Detailed study characteristics of eleven studies on **observational learning** (continued)

Reference	Population	Intervention and task	Amount of supervised practice	Measurement instruments, moments, and outcome
Mezzarobba et al., 2018 & 2020	N = 22 persons with PD	Task: Gait-related movements to circumvent freezing of gait.	All participants were instructed not to practice further rehabilitation/physiotherapy treatments.	Measurement instruments: 2018 - New Freezing of Gait Questionnaire (NFOGQ) - Unified Parkinson's Disease Rating Scale (UPDRS) II - Unified Parkinson's Disease Rating Scale (UPDRS) III - Modified Parkinson's Activity Scale - Berg Balance Scale (BBS) - 6-min Walking Test - Timed Up and Go - 39-item PD Questionnaire (PDQ39) – mobility - 39-item PD Questionnaire – activities of daily living - 39-item PD Questionnaire – emotional well-being - 39-item PD Questionnaire – stigma - 39-item PD Questionnaire – social support - 39-item PD Questionnaire – cognitions - 39-item PD Questionnaire – communication - 39-item PD Questionnaire – bodily discomfort - 39-item PD Questionnaire – total - Centre of Mass (COM) - Centre of Pressure (COP) – time - Centre of Pressure (COP) – position (all during sit-to-walk task) Measuring moments: Pre- and post-intervention, 4-week and 12-week follow-up Between-group differences: Between-group differences were found in favour of the action observation group at all three measurement moments for NFOGQ, UPDRS III, PDQ39 – mobility, and PDQ39 – bodily discomfort. Between-group differences were also found in favour of the action observation group for PDQ39 – total and UPDRS II (4 and 12 weeks), BBS (4 weeks), and the 6-min walking test (12 weeks). Significant differences were found in the COP profiles.
	Action observation plus sonification: *n* = 12; 74.7 yrs (SD: 5.9)	Action observation plus sonification was applied. Participants watched eight videos with eight different gait-related movements (e.g. weight shifting). Each video was composed of images and sounds (obtained with sonification). Afterwards, patients performed the actions for the same time twice (once without watching and once while watching again).	16 sessions of 60 min in 8 weeks	
	Cue control: *n* = 10; 72 yrs (SD: 5.8)	The same eight motor gestures were performed also in the cue control group with the same order and amount of time by using attentional strategies (e.g. stripes on the floor on beat of the metronome).	16 sessions of 60 min in 8 weeks	

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Table 6. Detailed study characteristics of eleven studies on **observational learning** (continued)

Reference	Population	Intervention and task	Amount of supervised practice	Measurement instruments, moments, and outcome
Pelosin et al., 2010	N = 18 persons with PD	Task: Strategies useful in circumventing freezing of gait.		Measurement instruments: - Freezing of Gait Questionnaire (FOG-Q) - Freezing of Gait (FOG) diary (number of FOG episodes, three conditions: start, turn, and obstacle) -Timed Up and Go Test -10-Metre Walking Test - Tinetti Scale - 39-item PD questionnaire Measuring moments: Pre- and post-intervention, 4-week follow-up Between-group differences: Between-group differences were found in favour of the action observation group for the FOG-Q (at follow-up) and the number of FOG episodes at follow-up (overall and for the start condition).
	Action observation: *n* = 9; 68.8 (SD: 4.1)	Participants watched six 6-min video clips twice each showing strategies to prevent FOG. After observation, patients were asked to practice (for the remaining time of the session – 36 minutes).	12 sessions of 60 min in 4 weeks	
	Landscape observation: *n* = 9; 70.2 (SD: 6.8)	Participants watched videos with landscape pictures for the same amount of time. They performed the same movement as the experimental group.	12 sessions of 60 min in 4 weeks	
Pelosin et al., 2018	N = 64 persons with PD	Task: Strategies useful in circumventing freezing of gait (FOG).		Measurement instruments: - A new FOG questionnaire - Timed Up and Go Test - 10-Metre Walking Test - Berg Balance Scale Measuring moments: Pre- and post-intervention, 4-weeks follow-up Between-group differences: Between-group differences were not reported.
	Action observation: *n* = 32; 70.4 (SD: 4.5)	Participants watched six 6-min video clips twice each showing strategies to prevent FOG. After observation, patients were asked to practice (for the remaining time of the session – 36 minutes).	10 sessions of 45 min in 5 weeks	
	Landscape observation: *n* = 32; 72.8 (SD: 3.1)	Participants watched videos with landscape pictures for the same amount of time. They performed the same movement as the experimental group.	10 sessions of 45 min in 5 weeks	

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Table 6. Detailed study characteristics of eleven studies on **observational learning** (continued)

Reference	Population	Intervention and task	Amount of supervised practice	Measurement instruments, moments, and outcome
Franceschini et al., 2012	N = 102 persons after stroke in the subacute phase	Task: 20 different daily routine tasks carried out with the upper limb (e.g. drinking from a glass, combing hair).	All participants underwent inpatient rehabilitation with at least 3 hours/day of physiotherapy.	Measurement instruments: - Frenchay Arm Test - Fugl-Meyer Test - Box and Block Test - Ashworth Scale – Shoulder - Ashworth Scale – Elbow - Ashworth Scale – Wrist - Functional Independence Measure – Motor Score Measurement moments: Pre- and post-intervention, 4-5 months follow-up Between-group differences: Differences in favour of the action observation group were found on Box and Block Test post-intervention and at follow-up.
	Effect of action observation: *n* = 53; 67.0 yrs (SD: 12.4)	Participants watched a video of one of the 20 tasks and imitated the action under supervision and with verbal feedback and assistance if needed.	20 sessions of 2 x 15 min in 4 weeks	
	'Sham' observation: *n* = 49; 65.7 yrs (SD: 11.9)	Participants were shown 5 static images displaying objects and executed a cognitive task. They performed limb movements according to a standard sequence, simulating those performed by the EG, involving shoulder and elbow joint mobilisation.	20 sessions of 2 x 15 min in 4 weeks	
Mancuso et al., 2021	N = 36 persons with stroke in acute phase	Task: Functional activities with the upper limbs. All participants received conventional therapy with a range of different patient-tailored interventions, not involving AOT or TOT objects.	All subjects underwent rehabilitative treatment for 4 consecutive weeks, 5 days per week. Each session consisted of 60 min.	Measurement instruments: - Fugl-Meyer Assessment for Upper Extremity - Box and Block Test - Functional Independence Measure (FIM) - Modified Ashworth Scale Measurement moments: Pre- and post-intervention Between-group differences: Differences in favour of the action observation group were found for the FIM.
	Action observation training (AOT): *n* = 16; median: 64.5 yrs (IQR: 15.8) (mean/SD: NR)	Participants observed 20 different videos of daily activities carried out with the upper limb and performed the activities afterwards (3 min of observation and 2 min of action performance for 3 motor sequences, repeated twice).	20 sessions of 30 min for 4 wks (additional to the rehabilitative treatment)	
	Task-oriented training (TOT): *n* = 16; median: 76.5 yrs (IQR: 13.7) (mean/SD: NR)	Participants performed the same functional activities with the upper limbs, in both unimanual and bimanual modalities, without watching the video beforehand.	20 sessions of 30 min for 4 wks (additional to the rehabilitative treatment)	

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Table 6. Detailed study characteristics of eleven studies on **observational learning** (continued)

Reference	Population	Intervention and task	Amount of supervised practice	Measurement instruments, moments, and outcome
Sale et al., 2014	N = 67 persons after stroke in the subacute phase 66.5 yrs (SD: 12.7)	Task: Functional upper limb activities.	All subjects underwent in-patient rehabilitation consisting of at least 3 hours/day	Measurement instruments: - Fugl-Meyer Test (FMT) – all subjects - Fugl-Meyer Test – right hemiparetic subjects - Fugl-Meyer Test – left hemiparetic subjects - Box and Block Test (BBT)-all subjects - Box and Block Test – right hemiparetic subjects - Box and Block Test – left hemiparetic subjects Measurement moments: Pre- and post-intervention, and 4-5 months follow-up Between-group differences: Between-group differences were found in favour of the action observation group for FM and BBT (post-intervention and follow-up, all subjects and left hemiparetic subjects).
	Action observation training: *n* = 33; data on age NR	Participants watched one of 20 videos with different routine actions carried out with the upper limb and performed the same movement afterwards.	20 sessions of 2x15 mins for 4 weeks	
	Control group: *n* = 34; data on age NR	Participants watched static images of objects and performed a cognitive task to keep their attention high. Afterwards, they performed the same movements as in the EC.	20 sessions of 2x15 mins for 4 weeks	
Rojasav et al., 2020	N = 33 elderly persons with mild cognitive impairment	Task: Walking.		Measurement instruments: - Gait speed (m/s) - Stride time variability (% coefficient of variation) - Stride length variability (% coefficient of variation) in both single- and dual-task conditions Measurement moments: Pre- and post-intervention, 4 wks follow-up Between-group differences: Between-group differences between the AOGT and CT groups were found in favour of the AOGT group for gait speed during single-task and dual-task conditions post-intervention and at follow-up.
	Action observation with gait training (AOGT): *n* = 11; 67.6 yrs (SD:4.6)	Participants watched a video of walking acted out by a normal healthy individual followed by gait training.	12 sessions of 65 mins for 4-6 weeks	
	Gait training (GT): *n* = 11; 67.5 yrs (SD: 5.6)	Participants watched a video of abstract pictures followed by the same gait training as in the experimental group.	20 sessions of 15 mins for 4-6 weeks	
	Control group (CT): *n* = 11; 65.7 yrs (SD: 2.5)	Participants received no training programme but were educated about dementia on the screening day.	1 session	

N = number of participants in total, n = number of participants per group, SD = standard deviation, NA = not applicable, NR = not reported, FIM = Functional Independence Measure, AFG = absolute functional gain, FOG = Freezing Of Gait, UPDR = Unified Parkinson's Disease Rating Scale, PDQ = Parkinson's Disease Questionnaire, NFOGQ = New Freezing of Gait Questionnaire, COP = Centre of Pressure, BBS = Berg Balance Scale, FOG-Q = Freezing of Gait Questionnaire, FIM = Functional Independence Measure, FMT = Fugl-Meyer Test, BBT = Box and Block Test, AOGT: = Action observation with gait training group, CT= control group

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning**

STROKE				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
An et al., 2021	N = 30 persons after stroke in sub-acute phase	Task trained in experimental group: Activities of daily living such as climbing stairs, making tea, folding tops or bottoms.	All participants received 10 min of occupational therapy	Measurement instruments: - Manual Function Test (MFT) Measurement moments: Pre- and post-intervention Between-group differences: Differences were found in the MFT in favour of the CT.
	Dual-task training group (DTG): *n* = 15; 65.2 yrs (SD: 12.2)	The secondary task included: • Attention tasks: subtracting or counting numbers, reading words backwards, or simple word games. • Executive function tasks: reacting to a virtual situation, explaining the order of wearing clothes, talking about daily routines, making a shopping list, naming certain categories, e.g. types of drinks.	25 sessions of 20 min in 5 weeks	
	Single-task training: *n* = 15; 65.3 yrs (SD: 12.7)	Sensory stimulation training (paralysed side), upper extremity muscle strength training, cognitive and perceptual training, and fine motor skill training.	25 sessions of 20 min in 5 weeks	
Fishbein et al., 2019	N = 22 persons after stroke in chronic phase	Task trained in experimental group: Walking All participants trained on a treadmill.		Measurement instruments: - Timed Up and Go Test - 10-Metre Walk Test - Functional Reach Test - Lateral Reach Test - Activity-Specific Balance Confidence - Berg Balance Scale Measurement moments: Pre- and post-intervention, follow-up Between-group differences: No significant differences in any of the outcomes.
	Dual-task walking group: *n* = 11; 64.4 yrs (SD: NR)	Walking while training with three VR games (ballgame, reactive boxing, and cleaning windows).	8 sessions of 20 min in 4 weeks	
	Single-task walking group: *n* = 11; 66 yrs (SD: NR)	Participants walked on a treadmill at a set speed.	8 sessions of 20 min in 4 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

STROKE				
Reference	**Population**	**Intervention and Task**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Meester et al., 2019	N = 50 persons after stroke in chronic phase	Task trained in experimental group: Walking. All participants trained on a treadmill.		Measurement instruments: walking distance was measured under single- and dual-task walking. Step activity was measured with a StepWatch Activity Monitor™: - Walking distance - Dual tasking walking distance change (effect) - Dual tasking walking distance and cognitive responses - Step activity per day Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training group (DTG): *n* = 26; 60.9 yrs (SD: 14.9)	Secondary tasks included listening or talking tasks: auditory Stroop task, serial subtraction, clock face task, letter fluency task, alternative uses, creativity task, radio (listening), planning of daily life activities.	20 sessions of 45 min in 10 weeks	
	Single-task training: *n* = 24; 62.3 yrs (SD: 15.5)	Participants were asked to walk with a focus on walking, with as little distraction as possible.	20 sessions of 45 min in 10 weeks	
ORTHOPAEDIC				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Conradsson & Halvarson, 2019	N = 68 older adults with osteoporosis	Task trained in experimental group: Balance training programme.		Measurement instruments: Gait parameters were measured on the GAITRite walkway system under single- and dual-task walking: - Step velocity - Step length - Cadence - Swing time - Step width - Step length asymmetry - Step time - Stance time - Step time asymmetry - Swing time asymmetry - Stance time asymmetry Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for cadence, swing time, and swing time asymmetry during normal gait and for step velocity, cadence, swing time, step time, swing time,
	Dual-task training group (DTG): *n* = 43; 76.0 yrs (SD: 6)	Each session included seated, standing, and walking exercises targeting aspects of postural control. Concurrent motor and cognitive tasks were performed (e.g. counting, carrying objects).	36 sessions of 45 min in 12 weeks	
	Control group: *n* = 25; 76.0 yrs (SD: 5)	Participants were encouraged to maintain their normal physical activities and were not restricted from participating in ongoing training regimens.	NA	

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				stance time, and step time variability during dual-tasking gait post-intervention in favour of the DTG.

.....Column Break..... **Table 7.** Detailed study characteristics of 43 studies on **dual-task learning** (continued)

ORTHOPAEDIC				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Konak et al., 2016	N = 42 older adults with osteoporosis	Task trained in experimental group: Exercise programmes on static balance, dynamic balance, and activity-specific balance confidence.		Measurement instruments: - One Leg Stance - Sports Kinesthetic Ability Trainer 4000 - Timed Up and Go - Berg Balance Scale (BBS) - Gait speed (in single- and dual-task conditions) Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for the BBS and walking speed in favour of the DTBG.
	Dual-task balance training group (DTBG): *n* = 22; 67.9 yrs (SD: 12.5)	Balance exercises (e.g. semi-tandem stance, one-leg stance, circle turns, or toe stance) under dual-task conditions (e.g. counting backwards, counting the days of the week, or naming objects).	12 sessions of 45 min in 4 weeks	
	Single-task balance training group: *n* = 20; 68.8 yrs (SD: 10.1)	Balance exercises (e.g. semi-tandem stance, one-leg stance, circle turns, or toe stance) without a secondary task.	12 sessions of 45 min in 4 weeks	
Uzunkulaoğlu et al., 2020	N = 50 older adults with osteoporosis	Task trained in experimental group: Balance training programme.		Measurement instruments: - Sports Kinesthetic Ability Trainer 2000 - Timed Up and Go - Berg Balance Scale (BBS) - Walking speed (in single- and dual-task conditions) - Activities-Specific Balance Confidence Scale Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task balance training group: *n* = 25; 72.3 yrs (SD: 5.5)	Balance exercises (e.g. tandem stance, semi-tandem stance, one- and two-legged stance, tandem walk, circle turns, heels and toes stance) were combined with a simultaneous cognitive task (e.g. singing a song, counting backwards from 10, and counting the days of the week).	12 sessions of 45 min in 4 weeks	
	Single-task balance training group: *n* = 25; 72.6 yrs (SD: 5.6)	Balance exercises (e.g. tandem stance, semi-tandem stance, one- and two-legged stance, tandem walk, circle turns, heel and toe stance) without a secondary task.	12 sessions of 45 min in 4 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

DEMENTIA				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Chen et al., 2018	N = 28 persons with mild-to-moderate dementia	Task trained in experimental group: Walking (forward or stepped sideways).		Measurement instruments: Gait parameters in three conditions (two dual tasks and one single task) - Walking speed - Stride length - DTC - Timed Up and Go Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Musical dual-task training group: *n* = 15; 77.3 yrs (SD: 9.4)	Walking while responding to obstacles (visual stimuli) and engaging in conversation (auditory stimuli). Other secondary tasks included singing or playing a (simple percussive) musical instrument.	8 sessions of 60 min in 8 weeks	
	Single-tasking condition: *n* = 13; 77.3 yrs (SD: 10.0)	Activities involving 1) non-musical cognitive tasks and 2) walking exercises (single-task conditions)	8 sessions of 60 min in 8 weeks	
Ghadiri et al., 2022	N = 38 older adults with dementia	Task trained in experimental group: Various locomotor tasks and manipulative skills (e.g. walking, running, jumping, hopping, galloping, catching, and throwing) in different displacement directions (e.g. forwards, backwards, left, right, straight, curvy, and diagonal).		Measurement instruments: Gait parameters in two conditions (single- and dual-task) - Walking speed - Stride length - Cadence Dual-task costs were calculated for - walking speed - stride length - cadence Measurement moments: Pre- and post-intervention Between-group differences: Difference was found in stride length in favour of the IDI compared to the DTI. Differences in cadence costs were found in favour of the IDI as evidenced by higher percentage change (reduction of cadence costs) compared to the DTI group.
	Dual-task intervention group (DTI): *n* = 19; 72.5 yrs (SD: 6.0)	Fundamental movement skills with cognitive activities, and locomotor and manipulative skills while completing a cognitive task (i.e. memory, attention, language, and executive function) were trained.	30 sessions of 50 min in 10 weeks	
	Iranian dance group (IDI): *n* = 19; 73 yrs (SD: 6.5)	Dance exercises chosen from the solo improvisation and folk categories of Iranian dance.	30 sessions of 50 min in 10 weeks	

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Table 10. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

DEMENTIA				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Lemke et al., 2018	N = 105 persons with dementia	Task trained in experimental group: Walking, sit-to-stand, and balance manoeuvres. Both groups received similar dementia-specific, patient-centred therapy sessions.		Measurement instruments: Gait parameters in 5 different conditions: - Gait speed - Cadence - Stride length - Single support - Dual-task costs In total, 55 outcomes related to motor performance were measured; see article for details. Measurement moments: Pre- and post-intervention and at 3-month follow-up Between-group differences: Differences in task performance between groups were observed for various outcomes. For all outcomes, the differences were always in favour of the dual task in comparison to the control condition.
	Dual-task training group: *n* = 56; 82.7 yrs (SD: 6.2)	Walking and counting in low- and high-demand dual-task conditions (2-forward and 3-backward calculations). Walk and calculate as fast as possible without prioritising one task.	20 sessions of 90 min in 10 weeks	
	Single-task training group: *n* = 49; 82.6 yrs (SD: 5.8)	Supervised motor placebo group training, including unspecific, low-intensity strength training and flexibility exercises for the upper body while seated.	20 sessions of 90 min in 10 weeks	
Menengiç et al., 2022	N = 20 persons with early-middle-stage Alzheimer's disease	Task trained in experimental group: Simple chair-based exercises.		Measurement instruments: - Five Times Sit to Stand Test (5XSST) - Timed Up and Go Test (TUG) - One-Leg Stance Test (for the left and right leg) - Katz Activities of Daily Living Scale (KATZ) - Functional Independence Measurement (FIM) Measurement moments: Pre- and post-intervention Between-group differences: Differences were found in 5XSST, TUG, KATZ, and FIM in favour of the motor-cognitive dual-tasking via TR group.
	Online dual-task exercise group via telerehabilitation (TR): *n* = 10; 77.7 yrs (SD: 5.3)	Real-time supervised motor-cognitive dual-task exercise treatment. Cognitive tasks were added to the physical exercises. Difficulty of cognitive tasks gradually increased each week.	25 sessions of 30 min in 6 weeks	
	Control group (CG): *n* = 10; 80.6 yrs (SD: 6.11)	No physical or cognitive intervention was given.	NA	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

DEMENTIA				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Schwenk et al., 2010	N = 61 persons with mild to moderate dementia	Task trained in experimental group: Walking, stepping, sitting, and balance training in static and dynamic conditions.		Measurement instruments: Gait performance in two conditions (serial 2-forward calculation and serial 3-backward calculation): - Gait speed - Cadence - Stride time - Stride length - Single support - Dual-task costs (DTC: only for this outcome were between-group differences calculated) Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for DTC (within the 3-backward calculation condition) in favour of the SDTG. For the other outcomes, no between-group differences were reported.
	Specific dual-task training group (SDTG): *n* = 26; 80.4 yrs (SD: 7.1)	Specific dual-task training and additional progressive resistance-balance and functional-balance training. Secondary tasks included motor (e.g. throwing or catching a ball) or cognitive (e.g. arithmetic, repeating names of animals) tasks.	24 sessions of 60 min in 12 weeks	
	Single-task training group: *n* = 35; 82.3 yrs (SD: 7.9)	Unspecific low-intensity exercise activities including flexibility exercise, calisthenics, and ball games while seated.	24 sessions of 60 min in 12 weeks	
PARKINSON'S				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Fernandes et al., 2015	N = 15 persons with Parkinson's	Task trained in experimental group: Balance training.		Measurement instruments: Postural sway was measured on a force platform under eyes-open and -closed conditions. - Postural sway (anteroposterior and mediolateral) - Timed Up and Go Test Measurement moments: Pre- and post-intervention

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PARKINSON'S				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Fernandes et al., 2015 (continued)	Dual-task training group (DTG): *n* = 7; 63.4 yrs (SD: 9.5 yrs)	Participants performed motor tasks (e.g. walking backwards while holding a basket) while they performed cognitive tasks (e.g. spelling words).	12 sessions of 60 min in 6 weeks	Between-group differences: Differences were found in anteroposterior and mediolateral sway post-intervention in favour of the DTG.
	Single-task training group: *n* = 8; 62.3 yrs (SD: 12.9 yrs)	Participants performed the same exercise as the dual-task training group minus the secondary cognitive task.	12 sessions of 60 min in 6 weeks	
Geroin et al., 2018	N = 121 persons with Parkinson's	Task trained in experimental group: Gait and functional training.	Both groups also performed 30 minutes of unsupervised training twice a week	Measurement instruments: Gait parameters were measured on the GAITRite walkway system under Auditory Stroop Task and Digit Span Task conditions: - Stride length - Swing time - Stride length SD - Stance % - Cadence - Stance time - Stride time - Double support % - Stride time SD - Support base - Swing % - Step length asymmetry Other outcomes: - Fall rate - Activities-Specific Balance Confidence Scale Measurement moments: Pre-, midterm, and post-intervention and at follow-up Between-group differences: Differences were found for stride length, cadence, stride time, swing %, stance %, stance time, double support % post-intervention and in the follow-up during dual-task walking in favour of IDTG.
	Integrated dual-task training (IDTG): *n* = 56; 65.8 yrs (SD: 9.2 yrs)	Participants performed gait training (e.g. turning and manoeuvring around the house) in combination with cognitive tasks (e.g. verbal fluency).	24 sessions of 40 min in 12 weeks	
	Consecutive task training: *n* = 65; 66.1 yrs (SD: 9.3 yrs)	Participants only practised single tasks. First, walking was practised; then, the cognitive task was practised.	24 sessions of 40 min in 12 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued).

PARKINSON'S				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Jäggi et al., 2023	N = 40 persons with Parkinson's	Task trained in experimental group: Balance and coordination. All participants received a conventional rehabilitation programme.		Measurement instruments: - Go/No Go Test - Gait speed - Short physical performance battery - Timed Up and Go Test Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training: *n* = 19; 71.9 yrs (SD: 9.1)	Participants trained on the exergaming device Dividat Senso. The training contained motor tasks (e.g. stepping in four directions) with simultaneous cognitive tasks (e.g. attentional focus).	15 sessions of 15 min in 3 weeks	
	Single-task training: *n* = 21; 72.9 yrs (SD: 10.1)	Participants underwent the conventional rehabilitation programme.	NR	
Silva et al., 2019, 2023	N = 25 persons with Parkinson's	Task trained in experimental group: Aquatic exercise.		Measurement instruments: - Timed Up and Go Test (TUG) - Five Times Sit to Stand Test (FTSTS) - Berg Balance Scale (BBS) - Dynamic Gait Index (DGI) - Unified Parkinson's Disease Rating Scale (UPDRS) - ADL - motor examination Measurement moments: Pre- and post-intervention, and at follow-up Between-group differences: In Silva et al. (2019), differences were found for the TUG, FTSTS, BBS, and DGI post-intervention and at follow-up in favour of the DTG. In Silva et al. (2023), differences for the UPDRS were NR.
	Dual-task training (DTG): *n* = 14; 63.1 yrs (SD: 13.6)	Motor tasks (e.g. aquatic walking exercises or passing a ball in a group) were combined with cognitive tasks (e.g. responding to whistle blows).	20 sessions of 60 min in 10 weeks	
	Control group: *n* = 11; 64.2 yrs (SD: 13.5)	Continuation of daily living.	NA	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

PARKINSON'S				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Valenzuela et al., 2020	N = 40 persons with Parkinson's	Task trained in experimental group: Gait training.		Measurement instruments: Gait parameters were measured under single-tasking and four different dual-tasking conditions (visual, verbal, auditory, and motor): - Gait velocity - Cadence - Stride length - Step width Measurement moments: Pre- and post-intervention, and at follow-up Between-group differences: Differences were found for gait velocity and stride length in all conditions, and single-tasking cadence post-intervention in the follow-up, in favour of the DTG.
	Dual-task training (DTG): *n* = 23; 66.4 yrs (SD: 7.1)	Participants performed gait training while performing a secondary cognitive (e.g. answering a complex question) or motor task (e.g. removing an item from your pocket).	20 sessions of 60 min in 10 weeks	
	Single-task training: *n* = 17; 64.8 yrs (SD: 8.8)	Gait training without secondary task.	20 sessions of 60 min in 10 weeks	
Yang et al., 2019	N = 18 persons with Parkinson's	Task trained in experimental group: walking forwards, obstacle crossing walking, walking on an S-shaped route, tandem walking, and walking backwards. Participants were progressively trained with increased difficulty of tasks.		Measurement instruments: Gait parameters in 3 different conditions (two dual-task and one single-task): - Walking speed - Double support time - Stride length - Stride time variability - Cadence - Dual-task cost-speed - Timed Up and Go Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for double support time in the cognitive dual-task condition in favour of the CDTT in comparison to the MDTT and CG. Differences were also found for stride time variability during motor dual-tasking in favour of the MDTT compared to the CDTT and CG.
	Cognitive dual-task gait training group (CDTT): *n* = 6; 65.0 yrs (SD: NR)	Cognitive dual tasks: repeating words, counting a 3-digit number, answering simple questions 'yes' or 'no', reciting a shopping list, reciting a short sentence backwards, and singing.	12 sessions of 30 min in 4 weeks	
	Motor dual-task gait training group (MDTT): *n* = 6; 69.5 yrs (SD: NR)	Motor dual tasks: holding one ball with both hands; bouncing a basketball with both hands; bouncing a basketball with either hand; bouncing one basketball with one hand and concurrently holding another basketball with the other hand.	12 sessions of 30 min in 4 weeks	
	Control training group (CG): *n* = 6; 66.5 yrs (SD: NR)	Practised the task without secondary tasks.	12 sessions of 30 min in 4 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Brustio et al., 2017	N = 60 older adults	Task trained in experimental group: Physical training based on balance and walking exercises. Motor tasks based around activities of daily living.		Measurement instruments: - Timed Up and Go Test - 6-Minute Walk Test - Four Square Step Test Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training group: *n* = 19; 74.3 yrs (SD: 2.6)	Holding progressively more difficult static postures or walking on a circular route, in a straight line, forwards, and backwards, was combined with ADL motor tasks (e.g. unbuttoning a shirt or unscrewing a nut and bolt).	32 sessions of 60 min in 16 weeks	
	Single-task training group: *n* = 19; 75.2 yrs (SD: 3.4)	The same exercises were performed as in the dual-tasking group minus the concurrent ADL motor tasks.	32 sessions of 60 min in 16 weeks	
	Control group: *n* = 22; 74.0 yrs (SD: 3.2)	Maintenance of their usual lifestyle without additional training.	NA	
Pessoa et al., 2020	N = 30 older adults	Task trained in experimental group: A variety of walking and balancing exercises.		Measurement instruments: Measurements were performed under single- and motor and cognitive dual-tasking: - Figure of 8 Walk (F8W) - Timed-up and Go Test (TUG) Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for TUG, TUG with concurrent motor task, and F8W with concurrent cognitive task in favour of the DTG.
	Dual-task training group (DTG): *n* = 15; 70.0 yrs (SD: NR)	Nine different motor-motor or motor-cognitive tasks were practised (e.g. walking the figure of eight while passing a ball around the waist, putting beans in a container with the non-dominant hand while singing a song, or throwing a ball against the wall and picking it up while reciting the days of the week backwards)	1 session of 40 min	
	Single-task training group: *n* = 15; 70.0 yrs (SD: NR)	The same exercises were performed as in the dual-tasking group minus the concurrent motor or cognitive task.	1 session of 40 min	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Gregory et al., 2016	N = 44 older adults	Task trained in experimental group: Group-based exercise programme with aerobic exercise, strength, balance, and flexibility training.		Measurement instruments: Gait parameters were measured on the GAITRite walkway system under single- and dual-task walking: - Gait velocity - Step length - Stride time Measurement moments: Pre- and post-intervention, and at follow-up Between-group differences: Differences were found for gait velocity, step length, and stride time variability during dual-task walking post-intervention and in the follow-up in favour of the DTG.
	Dual-task training group (DTG): *n* = 23; 72.6 yrs (SD: 7.0)	Participants performed square stepping exercises (following a pattern of forward, lateral, and diagonal foot placements on a mat) while answering cognitive challenging questions (e.g. semantic and phonemic verbal fluency task and 2- or 3-digit subtraction).	52 to 78 sessions of 75 min in 26 weeks	
	Exercise only group: *n* = 21; 74.5 yrs (SD: 7.4)	Participants received the same exercises as the dual-tasking group minus the cognitive task.	52 to 78 sessions of 75 min in 26 weeks	
Hiyamizu et al., 2012	N = 36 older adults	Task trained in experimental group: Strength and balance training.		Measurement instruments: - Chair Stand Test - Functional Reach Test - Timed Up and Go Test Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training group: *n* = 17; yrs (SD: 5.1)	Balance training was carried out on a regular floor, a balance pad plus, and a balance beam. This was combined with concurrent cognitive tasks (e.g. four-function calculation of up to two-digit numbers, comparing drawings and naming differences, or naming words in a category).	24 sessions of 60 min in 12 weeks	
	Single-task training group: *n* = 19; yrs (SD: 4.4)	Balance and strength exercises without secondary tasks.	24 sessions of 60 min in 12 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Javadpour et al., 2022	N = 69 older adults	Task trained in experimental group: Balance training.		Measurement instruments: Harmonic ratio and walking speed measured in single- and dual-task: - Harmonic ratio (in three axes) - Walking speed Other outcomes: - Fullerton Advanced Balance Scale - Activities-Specific Balance Confidence Scale - Timed Up and Go Test Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for all outcomes post-intervention (except for single-task gait speed) in favour of the STG and DTG compared to the CG. Differences between the STG and DTG were NR.
	Dual-task training group (DTG): *n* = 23; 68.9 yrs (SD: 3.4)	Participants performed balance exercises (e.g. tandem stance, walking on a foam surface, or a lateral lunge) while simultaneously performing a cognitive task (e.g. naming cities, naming boys' or girls' names, or counting backwards by four).	18 sessions of 40-60 min in 6 weeks	
	Single-task training group (STG): *n* = 23; 67.7 yrs (SD: 2.4)	The same balance exercises as the dual-task training group were performed minus the secondary tasks.	18 sessions of 40-60 min in 6 weeks	
	Control group (CG): *n* = 23; 69.3 yrs (SD: 3.7)	Continuation of daily life.	NA	
Kitazawa et al., 2015	N = 60 older adults	Task trained in experimental group: Net step exercise programme.		Measurement instruments: - Timed Up and Go Test Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training group: *n* = 30; 76.8 yrs (SD: 4.4)	Stepping exercises on a Fumamat (a grid made out of ropes) that started with a trial to get acquainted with the steps. Then, participants were instructed to follow the rhythm and direction of others while simultaneously singing children's songs.	8 sessions of 60 min in 8 weeks	
	Control group: *n* = 30; 75.5 yrs (SD: 3.7)	Continuation of daily living.	NA	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Nascimento et al., 2023	N = 44 older adults	Task trained in experimental group: Walking and Balancing.		Measurement instruments: - Timed Up and Go Test - Manual Timed Up and Go Test - Cognitive Timed Up and Go Test - Postural Balance Test - Sit to Stand Test Measurement moments: Pre-intervention, midterm, and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training group: *n* = 22; 66.1 yrs (SD: 4.1)	Different walking exercises and balance exercises (e.g. standing with eyes closed, walking on a soft surface, or walking with big steps) were combined with cognitive exercises (e.g. spelling words, memorising 3-5 words as quickly as possible).	24 sessions of 60 min in 12 weeks	
	Education control group: *n* = 22; 66.3 yrs (SD: 4.0)	Participants in this group joined educational workshops organised by a multidisciplinary team.	24 sessions of 60 min in 12 weeks	
Norouzi et al., 2019	N = 60 older adults	Task trained in experimental group: Isokinetic resistance training.		Measurement instruments: - Berg Balance Scale (BBS) Measurement moments: Pre- and post-intervention, and at follow-up Between-group differences: Differences were found for the BBS both post-intervention and at follow-up in favour of the MMDTG compared to the MCDTG and CG.
	Motor-motor training group (MMDTG): *n* = 20; 68.3 yrs (SD: 4.1)	Isokinetic resistance training on a machine for the lower limbs while performing motor tasks (e.g. throwing, holding, or balancing objects).	12 sessions of 60-80 min in 4 weeks	
	Motor-cognitive training group (MCDTG): *n* = 20; 68.5 yrs (SD: 3.7)	Isokinetic resistance training on a machine for the lower limbs while performing cognitive tasks (e.g. remembering shapes and colours).	12 sessions of 60-80 min in 4 weeks	
	Control group: *n* = 20; 68.1 yrs (SD: 3.7)	Participants met in groups to discuss issues related to daily life and retirement and to explore coping strategies.	12 sessions of 60-80 min in 4 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Plummer-D'Amato et al., 2012	N = 17 older adults	Task trained in experimental group: Walking and balance exercises.		Measurement instruments: - Timed Up and Go Test - Gait speed - Activities-Specific Balance Confidence Scale - 6-Metre Obstacle Course Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training group: *n* = 10; 76.6 yrs (SD: 5.6)	Gait agility training and balance training (e.g. walking on foam, obstacle negotiation, or a rope ladder) while performing cognitive tasks (e.g. naming words starting with the letter P, backward recitation, or naming a recent shopping list).	4 sessions of 45 min in 4 weeks	
	Single-task training group: *n* = 7; 76.7 yrs (SD: 6.0)	Identical gait agility and balance training without the cognitive secondary task.	4 sessions of 45 min in 4 weeks	
Raichlen et al., 2020	N = 74 older adults	Task trained in experimental group: Stationary biking.		Measurement instruments: Gait parameters were measured with accelerometers: - Stride length - Stride length variability - Stride duration - Stride duration variability - Stride velocity - Stride velocity variability Measurement moments: Pre-intervention, midterm, and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Cognitive training group: *n* = 21; 66.4 yrs (SD: 3.9)	The main task of training was navigating through a maze on a screen with handheld controllers with added tasks completed along the way (e.g. verbal paired associates or the N-back).	36 sessions of 30-40 min in 12 weeks	
	Exercise training group: *n* = 19; 68.1 yrs (SD: 3.9)	Aerobic exercise training on a stationary recumbent bicycle at 50-80% of HRR.	36 sessions of 30-40 min in 12 weeks	
	Cognitive and exercise training group: *n* = 20; 67.7 yrs (SD: 4.7)	Simultaneous cognitive and aerobic exercise training. A combination of the cognitive and exercise training groups.	36 sessions of 30-40 min in 12 weeks	
	Control group: *n* = 14; 69.3 yrs (SD: 4.3)	Participants watched videos that were relatively neutral in terms of interest and content.	36 sessions of 30-40 min in 12 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued).

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Sinaei et al., 2016	N = 24 older adults	Task trained in experimental group: Balance training.		Measurement instruments: - Fullerton Advanced Balance Scale Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task training group: *n* = 12; 63.8 yrs (SD: 5.0)	Participants performed balance tasks (e.g. catching and throwing a ball in stance, controlling balance during the Romberg test, or balancing on a gym ball) while performing a cognitive task (e.g. naming streets, naming flowers, or counting backwards).	12 sessions of 45 min in 4 weeks	
	Single-task training group: *n* = 12; 65.2 yrs (SD: 4.9)	Participants performed the same exercises as the dual-tasking training group minus the cognitive task.	12 sessions of 45 min in 4 weeks	
Tasvuran Horata et al., 2021	N = 32 older adults	Task trained in experimental group: Walking and Balancing.		Measurement instruments: Gait parameters in single- and dual-tasking conditions: - Gait speed - Cadence - Step length (left and right) Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for gait speed in single- and dual-tasking, right step length in dual-tasking, and cadence in single- and dual-tasking post-intervention in favour of the DTG.
	Dual-task training group (DTG): *n* = 16; 65.6 yrs (SD: 2.6)	Participants performed gait and balance tasks (e.g. reaching, fast walking, or single-leg stance) while performing cognitive tasks (e.g. recalling a number sequence, counting forwards, or drawing a letter on the floor with your foot).	12 sessions of 60 min in 6 weeks	
	Single-task training group: *n* = 16; 64.6 yrs (SD: 3.3)	Participants performed the same motor exercises as the dual-task training group without the secondary tasks.	12 sessions of 60 min in 6 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued).

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Trombetti et al., 2011	N = 134 older adults	<u>Task trained in experimental group</u>: Walking and balancing. Both groups received dual-task training, but one from the start and the other after 6 months.		<u>Measurement instruments</u>: GAITRite walkway system during single- and dual-tasking: - Gait velocity - Support base - Stride length - Stride time variability - Cadence - Stride length variability - Double support phase Other outcomes: - Timed Up and Go Test - Tinetti Test - Fall rate
	<u>Early intervention group (EIG):</u> *n* = 66; 75.0 yrs (SD: 8.0)	Participants in this group performed walking and balancing exercises (e.g. multidirectional weight shifting, turn sequences, or exaggerated upper body movements while walking and standing). The secondary task was a musical cue (e.g. walking to the rhythm, playing a percussion instrument, or adapting to a change in rhythms).	26 sessions of 60 min in 26 weeks	
	<u>Delayed intervention control group:</u> *n* = 68; 76.0 yrs (SD: 6.0)	Subjects in the delayed control group were instructed to maintain their usual lifestyle for the first part of the study, whereafter they commenced the same intervention as the early intervention group.	26 sessions of 60 min in 26 weeks	<u>Measurement moments</u>: Pre- and post-intervention and at follow-up <u>Between-group differences:</u> Differences were found in fall rate, gait velocity, stride length, stride time variability during single-task walking post-intervention in favour of the EIG. Differences were also found in stride length and stride length variability during dual-task walking post-intervention in favour of the EIG. Differences were also found in the TUG and the Tinetti Test post-intervention in favour of the EIG.

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued).

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Uemura et al., 2012	N = 15 older adults	Task trained in experimental group: Weight shifting exercises, start and stop exercises, and walking exercises with directional shifts.	All participants received 30 min of seated group training once a week	Measurement instruments: - Gait initiation test - 10-metre steady-state gait test Measurement moments: Pre- and post-intervention Between-group differences: No significant differences in any of the outcomes.
	Dual-task switch group: *n* = 8; 82.4 yrs (SD: 5.9)	The secondary task included performing cognitive tasks (e.g. forward counting and reciting of letters from the Japanese alphabet, reciting as many things as possible from a category, and responding to an auditory cue as fast as possible). The focus of the STE was to improve the ability to initiate and switch movements quickly under a dual-task condition.	24 sessions of 5 min in 24 weeks	
	Control exercise group: *n* = 7; 82.4 yrs (SD: 6.8)	Walking exercises (e.g. backwards and sideways) were followed by cognitive exercises (e.g. reciting letters of the Japanese alphabet). The focus of the control group was to improve steady-state walking training under dual-task conditions.	24 sessions of 5 min in 24 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued).

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Wollesen et al., 2015	N = 38 older adults	Task trained in experimental group: Walking and balancing.		Measurement instruments: Gait measurements during single- and dual-tasking: - Step width - Step length (left and right) - Gait line (left and right) - Step length SD (left and right) - Step width COV - Step length COV (left and right) - Peak pressure forefoot (left and right) - Peak pressure midfoot (left and right) - Peak pressure heel (left and right) Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for gait line left and right (length of foot rolling movements) during single- and dual-task walking post-intervention in favour of the DTG. Differences were also found for midfoot peak pressure left during single-tasking and left and right during dual-tasking post-intervention in favour of the DTG.
	Dual-task training group (DTG): *n* = 19; 73.2 yrs (SD: NR)	Participants performed motor tasks (e.g. brisk walking, starting and stopping, and avoiding obstacles) with cognitive tasks (e.g. paying attention to tripping hazards or speed rating). In the second phase of the trial, difficulty was increased with e.g. time pressure and task prioritisation.	12 sessions of 60 min in 12 weeks	
	Control group: *n* = 19; 72.1 yrs (SD: NR)	Continuation of daily living.	NA	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued).

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Wollesen et al., 2017A	N = 78 older adults	Task trained in experimental group: Walking and balancing.		Measurement instruments: Gait measurements were made with a treadmill under single- and dual-tasking: - Step length (left and right) - Step width - Gait line (left and right) - Peak pressure – forefoot (left and right) - Peak pressure – midfoot (left and right) - Peak pressure – heel (left and right) Measurement moments: Pre- and post-intervention Between-group differences: Between-group differences were found for step length and gait line during single- and dual-task walking in favour of the BDT intervention compared to the SRG and CG.
	Balance and task managing training group (BDT): *n* = 29; 70.7 yrs (SD: 4.9)	Participants performed motor tasks (e.g. brisk walking, starting and stopping, and avoiding obstacles) together with cognitive tasks (e.g. paying attention to tripping hazards or speed rating). Also, task managing strategies were taught, such as, 'If you need to stop your walking, quickly bend your knees.'	12 sessions of 60 min in 12 weeks	
	Strength and resistance training group (SRG): *n* = 23; 71.7 yrs (SD: 4.9)	General strength training of major muscle groups.	12 sessions of 60 min in 12 weeks	
	Control group (CG): *n* = 26; 73.7 yrs (SD: 5.0)	Continuation of daily living.	NA	
Wollesen et al., 2017B	N = 95 older adults	Task trained in experimental group: Walking and balancing.		Measurement instruments: Gait measurements were made with a treadmill under single- and dual-tasking: - Step length (left and right) - Step width - Gait line (left and right) Other outcomes: - Short Physical Performance Battery - Stroop walking test
	Dual-task + no-fear group (DTG): *n* = 26; 72.2 yrs (SD: 4.6) Dual-task + fear group (DTGF): *n* = 30; 69.8 yrs (SD: 5.7)	Participants performed motor tasks (e.g. brisk walking, starting and stopping, and avoiding obstacles) with concurrent cognitive tasks (e.g. changing directions or following rhythms) which increased in difficulty over time. Participants were later subdivided into groups with a fear of falling and no fear of falling.	12 sessions of 60 min in 12 weeks	

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	Control + no-fear group: *n* = 19; 72.9 yrs (SD: 4.4) Control + fear group: *n* = 20; 72.7 yrs (SD: 5.3)	Continuation of daily living. Participants were later subdivided into groups with a fear of falling and no fear of falling.	NA	Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for step length in both feet and gait line in the right foot post-intervention in favour of the DTG groups.

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Yamada et al., 2011A	N = 84 older adults	Task trained in experimental group: DVD group training from a seated position. Stepping up and down, aerobic exercise with arm and leg movements targeting the major muscle groups, strength training with an elastic band around the leg, and balance training with weight shifting.		Measurement instruments: - 10-Metre Walking Test (10MWT) - Timed Up and Go Test - 5 Chair Stand Test Measurement moments: Pre- and post-intervention Between-group differences: Differences were found in the 10MWT during dual-task walking post-intervention in favour of the DTG.
	Dual-task training group (DTG): *n* = 41; 83.0 yrs (SD: 6.7)	During the stepping up and down exercise, participants performed cognitive tasks (e.g. naming animals, vegetables, and fruits, or words that start with an A).	48 sessions of 20 min in 24 weeks	
	Control group: *n* = 43; 82.9 yrs (SD: 5.5)	Continuation of daily living.	NA	
Yamada et al., 2011B	N = 50 older adults	Task trained in experimental group: DVD group training from a seated position. Stepping up and down, aerobic exercise with arm and leg movements targeting the major muscle groups, strength training with an elastic band around the leg, and balance training with weight shifting.		Measurement instruments: Gait parameters were measured under single-tasking and cognitive and motor dual-tasking: - Gait speed - 10 m walking step - 10 m walking cadence - Dual-task costs Other outcomes: - Timed Up and Go Test - Functional reach - Single-leg stance - Number of steps/5sec - Number of stepped figures/5sec Measurement moments: Pre- and post-intervention Between-group differences: NR
	Dual-task training group: *n* = 26; 80.3 yrs (SD: 5.4)	During the stepping up and down exercise, participants performed cognitive tasks (e.g. naming animals, vegetables, and fruits, or words that start with an A).	24 sessions of 50 min in 24 weeks	
	Single-task training group: *n* = 24; 81.2 yrs (SD: 7.6)	Participants performed the same exercise as the dual-task training group minus the secondary cognitive task.	24 sessions of 50 min in 24 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued).

OLDER ADULTS WITH BALANCE IMPAIRMENTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Azadian et al., 2016	N = 30 older adults with balance impairments	Task trained in experimental group: Gait training and balancing.		Measurement instruments: Gait analysis was measured with the Vicon system: - Cadence (steps/min) - Gait speed - Stride time - Stride length - Stance time - Step length - Swing time - Opposite foot off - Step time - Opposite foot contact - Single support - Foot off - Double support Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for cadence, stride time, stance time, swing time, and single support time in favour of the DTG and EFG compared to the CG. Differences were also found for gait speed, step time, and double support time in favour of the EFG compared to the DTG and CG.
	Dual-task training group (DTG): *n* = 10; 73.9 yrs (SD: 5.5 yrs)	Standing and walking exercises (e.g. weight shifting in stance) with a concurrent cognitive task (e.g. naming animals), the difficulty of which increased over time.	24 sessions of 45 min in 8 wks	
	Executive function training EFG): *n* = 10; 73.8 yrs (SD: 3.9 yrs)	A mix of 20 tasks involving working memory, and inhibitory and speed processing tasks.	24 sessions of 45 min in 8 wks	
	Control group (CG): *n* = 11; 73.7 yrs (SD: 4.4 yrs)	Continuation of daily living.	NA	
Khan et al., 2018	N = 39 older adults with balance impairments	Task trained in experimental group: Balance training, e.g. practising weight shifting, transfer, and static/dynamic standing.		Measurement instruments: - Functional Reach Test - Timed Up and Go Test (TUG) - Berg Balance Scale (BBS) - 10-Metre Walk Test Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for TUG and BBS post-intervention in favour of the TCG group.
	Dual-task training group: *n* = 19; 63 yrs (SD: NA)	The secondary motor task included holding a glass of water, throwing/catching a ball, placing pencils in holes of a covered cup, cardboard pin plugging, pegging a board in 2 different containers, transferring objects at different heights.	18 sessions of 40-50 min in 6 weeks	
	Turning and cognitive training group (TCG): *n* = 20; 63 yrs (SD: NA)	The secondary cognitive tasks included verbal fluency, sentence completion, forward/backward counting from 1-10, 3-serial subtraction, an auditory choice reaction task, and visuospatial tasks.	18 sessions of 40-50 min in 6 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS WITH BALANCE IMPAIRMENTS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Silsupadol et al., 2009 A&B	N = 21 older adults with balance impairments	Task trained in experimental group: Balance training.		Measurement instruments: Gait speed and stride length were measured for narrow walk and obstacle course in single- and dual-tasking: - Gait speed - Stride length Other outcomes: - Berg Balance Scale - Activities-Specific Balance Confidence Scale Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for dual-tasking gait speed post-intervention in favour of the DTVG and DTFG compared to the STG. Differences were also found for the dual-tasking narrow walk post-intervention in favour of the DTVG compared to the DTFG and STG.
	Dual-tasking variable priority group (DTVG): *n* = 6; 76.0 yrs (SD: 4.7)	Progressive balance exercise programme related to body stability and transport while performing cognitive tasks (e.g. remembering numbers). Focus was first put on the motor task and later on the cognitive task.	12 sessions of 45 min in 4 weeks	
	Dual-tasking fixed priority group (DTFG): *n* = 8; 74.4 yrs (SD: 6.2)	Progressive balance exercise programme related to body stability and transport while performing cognitive tasks (e.g. remembering numbers). Participants were instructed to maintain focus on both tasks.	12 sessions of 45 min in 4 weeks	
	Single-task training group (STG): *n* = 7; 74.7 yrs (SD: 7.8)	Progressive balance exercise programme related to body stability and transport.	12 sessions of 45 min in 4 weeks	
OLDER ADULTS WITH A HISTORY OF FALLS				
Reference	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Park et al., 2022	N = 58 older adults with a history of falls	Task trained in experimental group: Walking and balance training.		Measurement instruments: - One Leg Standing Test - Timed Up and Go Test Measurement moments: Pre- and post-intervention Between-group differences: Differences were found for all outcomes in favour of the DTG.
	Dual-task training group (DTG): *n* = 29; 71.8 yrs (SD: 3.2)	Motor tasks (e.g. alternatively placing a foot on a step, walking, or stepping over an obstacle) were combined with cognitive tasks (e.g. reciting days backwards, double-digit subtraction, or carrying a grocery bag).	12 sessions of 45 min in 6 wks	
	Balance training group: *n* = 29; 70.9 yrs (SD: 2.8)	Stability exercises (e.g. tandem stance, walking backwards, transfer from one chair to another) without secondary tasks.	12 sessions of 45 min in 6 wks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS WITH COGNITIVE IMPAIRMENTS				
References	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Delbroek et al., 2017	N = 60 older adults with mild cognitive impairment	Task trained in experimental group: Standing balance tasks including weight-bearing exercises on BioRescue platform.		Measurement instruments: - Performance-Oriented Mobility Assessment Instrumented Timed Up and Go Test during single-tasking: - Total duration - Turn: duration - Sit to stand: duration - Turn to sit: duration - Turn: step-time before turn iTUG during dual-tasking: - Turn: duration - Sit to stand: duration - Turn: step time before turn - Error rate Measurement moments: Pre- and post-intervention Between-group differences: NR
	Dual-task training (with the BioRescue) group: *n* = 10; 86.9 yrs (SD: 5.6)	Training programme with the BioRescue (RM Ingenierie, France). The secondary tasks included memory exercises (e.g. remembering cards) or avoiding objects.	12 sessions of 24 min in 6 weeks	
	Control group: *n* = 10; 87.5 yrs (SD: 6.6)	Continuation of usual care in the nursing home if applicable.	6 weeks (sessions NR)	
Kuo et al., 2022	N = 30 older adults with mild cognitive impairment	Task trained in experimental group: Walking.		Measurement instruments: Gait performance under single and under cognitive and motor dual-task conditions - Speed - Cadence - Stride length - Stride time - Spatial variability - Temporal variability - Dual-task costs (only within dual-task conditions) Measurement moments: Pre- and post-intervention and 4-week follow-up
	Cognitive dual-task training group (CDTT): *n* = 9; 80 yrs (SD: NA)	Cognitive tasks during walking on level surface. Cognitive tasks included repeating phrases, counting forwards or backwards, playing phonemic word chain games, reciting a poem, having conversations, and reciting a sentence backwards.	24 sessions of 45 min in 8 weeks	
	Motor dual-task training Group (MDTT): *n* = 11; 87 yrs (SD: NA)	Motor tasks during walking on level surface. Motor tasks included holding balls, raising an umbrella with both hands, waving a rattle, beating a castanet, bouncing a basketball, and holding one ball and concurrently kicking a basketball.	24 sessions of 45 min in 8 weeks	

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Table 7. Detailed study characteristics of 43 studies on **dual-task learning** (continued)

OLDER ADULTS WITH COGNITIVE IMPAIRMENTS				
References	**Population**	**Intervention**	**Amount of supervised practice**	**Measurement instruments, moments, and outcome**
Kuo et al., 2022 (continued)	Conventional physical therapy group (CPT): *n* = 10; 79 yrs (SD: NA)	A multicomponent exercise programme consisting of muscle strengthening, balance, and gait training.	24 sessions of 45 min in 8 weeks	Between-group differences: Differences were found for motor dual-task walking performance in favour of the MDTT and CDTT groups compared to the CPT group. Only the MDTT maintained this result in the follow-up.
Parial et al., 2022	N = 60 older adults with mild cognitive impairment	Task trained in experimental group: Zumba dance		Measurement instruments: - Short Physical Performance Battery Measurement moments: Pre- and post-intervention, and 6-week follow-up Between-group differences: NR
	Dual-task Zumba Gold: *n* = 30; 63.3 yrs (SD: 4.5)	Besides the Zumba Gold dancing routine, the following secondary tasks were performed: answering questions about the participants' orientation to a person (identifying their names), time (date, month, or year), and place (current location); forward and backward serial counting; doing arm-clock positions based on prompts; forward and backward recall of word/number series; and forward and backward spelling.	36 sessions of 60 min in 12 weeks	
	Control group: *n* = 30; 643 yrs (SD: 5.9)	Participants received health education on dementia risk reduction, which highlighted the importance of physical activities and lifestyle factors for preventing cognitive decline. Participants were instructed to perform moderate physical/leisure activities.	36 sessions of 60 min in 12 weeks	

N = number of participants in total, n = number of participants per group, *SD = standard deviation, NA = not applicable, NR = not reported, DTG = Dual-task training group; STG = Single-task training group, CG = control group, MFT = Manual Function Test, DTBG = Dual-Task Balance training Group, BBS = Berg Balance Scale, DTI = dual-task intervention group, IDI = Iranian dance group, 5XSST = Five Times Sit to Stand Test, TUG = Timed Up and Go Test, KATZ = Katz Activities of Daily Living Scale, FIM = Functional Independence Measurement, SDTG = Specific dual-task training group, DTC = Dual-task costs, IDTG = Integrated dual-task training group, FTSTS = Five Times Sit to Stand Test, DGI = Dynamic Gait Index, UPDR = Unified Parkinson's Disease Rating Scale, CDTT = Cognitive dual-task gait training group, MDTT = Motor dual-task gait training group, CTT= Control training group, F8W = Figure of 8 Walk, MMDTG = Motor-motor training group, MCDTG = Motor-cognitive training group, EIG = Early intervention group, 10MWT = 10-Metre Walking Test, EFG = Executive function training, DTVG = Dual-tasking variable priority group, DTFG = Dual-tasking fixed priority group, CDTT = Cognitive dual-task training group, MDTT = Motor dual-task training group, CDT = Conventional physical therapy group*

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